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FIXTURES FOR FLUOROSCOPIC IMAGING SYSTEMS AND RELATED NAVIGATION SYSTEMS AND METHODS

Abstract

A fixture for a fluoroscopic x-ray imaging system is discussed, wherein the fluoroscopic imaging system includes a C-arm, an x-ray source at a first end of the C-arm, and an x-ray detector at a second end of the C-arm. The fixture includes a processor and memory coupled with the processor. The memory includes instructions that are executable by the processor so that the processor is configured to detect an x-ray emission from the x-ray source toward the x-ray detector, determine an offset of the x-ray source relative to the x-ray detector responsive to detecting the x-ray emission, and provide an indication of the offset of the x-ray source to a medical navigation system. Related methods and robotic systems are also discussed.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. patent application Ser. No. 17/547,605, filed on Dec. 10, 2021 (published as U.S. Pat. Pub. No. 2022-0378388), which is a continuation application of U.S. patent application Ser. No. 16/860,125 filed on Apr. 28, 2020, each of which is incorporated in its entirety herein.

FIELD

[0002] The present disclosure relates to medical devices, and more particularly, fluoroscopic x-ray imaging systems and related methods and devices.

BACKGROUND

[0003] Prior to and/or during a surgical procedure performed using surgical navigation, registration between a coordinate system of a fluoroscopic x-ray imaging system, a coordinate system of a tracking system (e.g., a camera coordinate system), and/or a coordinate system of a pre-operative image of the anatomy (e.g., an MRI or CT scan) may be desired. Because the fluoroscopic imaging system is a large piece of equipment with an x-ray source and an x-ray detector supported at opposite ends of a large C-arm, the C-arm may flex differently in different positions so that relative positions of the x-ray source and the x-ray detector may be different at different positions of the C-arm, making it difficult to properly register resulting images with respect to other coordinate systems.

SUMMARY

[0004] According to some embodiments of inventive concepts, a surgical imaging system is configured for use with a fluoroscopic imaging system. The fluoroscopic imaging system includes a C-arm, an x-ray source at a first end of the C-arm, and an x-ray detector at a second end of the C-arm, and the fluoroscopic imaging system is configured to generate x-ray images based on x-rays received at the x-ray detector from the x-ray source. The surgical imaging system includes a fixture and a medical navigation system. The fixture includes an x-ray opaque fiducial pattern in a single plane, and the fixture is coupled with the x-ray detector so that the x-ray opaque fiducial pattern is on a surface of the x-ray detector between the x-ray detector and the x-ray emitter. The medical navigation system is configured to receive a first patient image from the fluoroscopic imaging system corresponding to a first orientation of the C-arm with the first patient image including first shadows corresponding to the x-ray opaque fiducial pattern. The medical navigation system is also configured to receive a second patient image from the fluoroscopic imaging system corresponding to a second orientation of the C-arm with the second patient image including second shadows corresponding to the x-ray opaque fiducial pattern. The medical navigation system is further configured to provide a registration between a tracking coordinate system for a physical space monitored by tracking cameras and an image coordinate system for the first patient image and the second patient image based on a correlation of offsets of the x-ray source relative to the x-ray detector as a function of an orientation of the x-ray detector relative to gravity and based on the first shadows in the first patient image and the second shadows in the second patient image.

[0005] According to some embodiments of inventive concepts, a fixture is provided for a fluoroscopic x-ray imaging system, wherein the fluoroscopic x-ray imaging system includes a C-arm, an x-ray source at a first end of the C-arm, and an x-ray detector at a second end of the C-arm. The fixture includes a processor and memory coupled with the processor. The memory includes instructions that are executable by the processor so that the fixture is configured to detect an x-ray

emission from the x-ray source toward the x-ray detector, determine an offset of the x-ray source relative to the x-ray detector responsive to detecting the x-ray emission, and provide an indication of the offset of the x-ray source to a medical navigation system.

[0006] According to some other embodiments of inventive concepts, a method is provided to operate a fixture for a fluoroscopic x-ray imaging system including a C-arm, an x-ray source at a first end of the C-arm, and an x-ray detector at a second end of the C-arm. An x-ray emission from the x-ray source toward the x-ray detector is detected. An offset of the x-ray source relative to the x-ray detector is determined responsive to detecting the x-ray emission. The offset of the x-ray source is provided to a medical navigation system.

[0007] According to still other embodiments of inventive concepts, a medical navigation system includes a processor and memory coupled with the processor wherein the memory includes instructions that are executable by the processor so that the medical navigation system is configured to perform the following operations. The processor is configured to receive information defining a first image from a fluoroscopic x-ray imaging system, wherein the fluoroscopic x-ray imaging system includes a C-arm, an x-ray source at a first end of the C-arm, and an x-ray detector at a second end of the C-arm. The processor is configured to provide/receive an indication of a first offset of the x-ray source relative to the x-ray detector, with the first offset being associated with the first image. The processor is configured to receive information defining a second image from the fluoroscopic x-ray imaging system, with the first image and the second image being different. The processor is configured to provide/receive an indication of a second offset of the x-ray source relative to x-ray detector, with the second offset being associated with the second image and with the first offset and the second offset being different. The processor is configured to provide a registration between a tracking coordinate system for a physical space monitored by tracking sensors and an image coordinate system for the first and second images from the fluoroscopic x-ray imaging system, with the registration being provided based on the indications of the first offset and the second offset.

[0008] According to yet other embodiments of inventive concepts, a method is provided to operate a medical navigation system. Information defining a first image is received from a fluoroscopic x-ray imaging system, with the fluoroscopic x-ray imaging system including a C-arm, an x-ray source at a first end of the C-arm, and an x-ray detector at a second end of the C-arm. An indication of a first offset of the x-ray source relative to the x-ray detector is provided/received with the first offset being associated with the first image. Information defining a second image is received from the fluoroscopic x-ray imaging system, with the first and second images being different. An indication of a second offset of the x-ray source relative to the x-ray detector is provided/received, with the second offset being associated with the second image and with the first offset and the second offset being different. A registration between a tracking coordinate system for a physical space monitored by tracking sensors and an image coordinate system for the first and second images from the fluoroscopic x-ray imaging system is provided based on the indications of the first and second offsets.

[0009] Other methods and related systems, and corresponding methods and computer program products according to embodiments of the inventive subject matter will be or become apparent to one with skill in the art upon review of the following drawings and detailed description. It is intended that all such systems, and corresponding methods and computer program products be included within this description, be within the scope of the present inventive subject matter and be protected by the accompanying claims. Moreover, it is intended that all embodiments disclosed herein can be implemented separately or combined in any way and/or combination.

Description

DESCRIPTION OF THE DRAWINGS

[0010] The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this application, illustrate certain non-limiting embodiments of inventive concepts. In the drawings:

[0011] FIG. 1 is an overhead view of a potential arrangement for locations of the robotic system, patient, surgeon, and other medical personnel during a surgical procedure;

[0012] FIG. 2 illustrates the robotic system including positioning of the surgical robot and the camera relative to the patient according to one embodiment;

[0013] FIG. 3 illustrates a surgical robotic system in accordance with an exemplary embodiment;

[0014] FIG. 4 illustrates a portion of a surgical robot in accordance with an exemplary embodiment;

[0015] FIG. 5 illustrates a block diagram of a surgical robot in accordance with an exemplary embodiment;

[0016] FIG. 6 illustrates a surgical robot in accordance with an exemplary embodiment;

[0017] FIGS. 7A-7C illustrate an end-effector in accordance with an exemplary embodiment;

[0018] FIG. 8 illustrates a surgical instrument and the end-effector, before and after, inserting the surgical instrument into the guide tube of the end-effector according to one embodiment;

[0019] FIGS. 9A-9C illustrate portions of an end-effector and robot arm in accordance with an exemplary embodiment;

[0020] FIG. 10 illustrates a dynamic reference array, an imaging array, and other components in accordance with an exemplary embodiment;

[0021] FIG. 11 illustrates a method of registration in accordance with an exemplary embodiment;

[0022] FIG. 12A-12B illustrate embodiments of imaging devices according to exemplary embodiments;

[0023] FIG. 13A illustrates a portion of a robot including the robot arm and an end-effector in accordance with an exemplary embodiment;

[0024] FIG. 13B is a close-up view of the end-effector, with a plurality of tracking markers rigidly affixed thereon, shown in FIG. 13A;

[0025] FIG. 13C is a tool or instrument with a plurality of tracking markers rigidly affixed thereon according to one embodiment;

[0026] FIG. 14A is an alternative version of an end-effector with moveable tracking markers in a first configuration;

[0027] FIG. 14B is the end-effector shown in FIG. 14A with the moveable tracking markers in a second configuration;

[0028] FIG. 14C shows the template of tracking markers in the first configuration from FIG. 14A;

[0029] FIG. 14D shows the template of tracking markers in the second configuration from FIG. 14B;

[0030] FIG. 15A shows an alternative version of the end-effector having only a single tracking marker affixed thereto;

[0031] FIG. 15B shows the end-effector of FIG. 15A with an instrument disposed through the guide tube;

[0032] FIG. 15C shows the end-effector of FIG. 15A with the instrument in two different positions, and the resulting logic to determine if the instrument is positioned within the guide tube or outside of the guide tube;

[0033] FIG. 15D shows the end-effector of FIG. 15A with the instrument in the guide tube at two different frames and its relative distance to the single tracking marker on the guide tube;

[0034] FIG. 15E shows the end-effector of FIG. 15A relative to a coordinate system;

[0035] FIG. 16 is a block diagram of a method for navigating and moving the end-effector of the robot to a desired target trajectory;

[0036] FIGS. **17A-17B** depict an instrument for inserting an expandable implant having fixed and moveable tracking markers in contracted and expanded positions, respectively;

[0037] FIGS. **18A-18B** depict an instrument for inserting an articulating implant having fixed and moveable tracking markers in insertion and angled positions, respectively;

[0038] FIG. **19A** depicts an embodiment of a robot with interchangeable or alternative end-effectors;

[0039] FIG. **19B** depicts an embodiment of a robot with an instrument style end-effector coupled thereto;

[0040] FIG. **20** is a photograph illustrating a fixture including multiple BB/fiducial planes;

[0041] FIG. **21** is a schematic diagram illustrating a pinhole camera model;

[0042] FIG. **22** is a diagram illustrating a configuration of a fluoroscopic x-ray imaging system according to some embodiments of inventive concepts;

[0043] FIGS. **23A, 23B, and 23C** are diagrams illustrating a gravity vector according to some embodiments of inventive concepts;

[0044] FIGS. **24A and 24B** are photographs illustrating a gravity vector according to some embodiments of inventive concepts;

[0045] FIGS. **25 and 26** are a flow charts illustrating calibration operations according to some embodiments of inventive concepts;

[0046] FIGS. **27A, 27B, and 27C** are photographs illustrating fixtures that may be used for calibration according to some embodiments of inventive concepts;

[0047] FIG. **28** is a photograph illustrating a reference fluoroscopic image/shot as discussed with respect to FIG. **25** according to some embodiments of inventive concepts;

[0048] FIGS. **29A, 29B, and 29C** are photographs illustrating use of BBs/fiducials in a fixture to provide top-bottom/left-right collimation according to some embodiments of inventive concepts;

[0049] FIG. **30** illustrates use of a crosshair as a fiducial according to some embodiments of inventive concepts;

[0050] FIG. **31** illustrates use of surveillance markers on a fluoroscopic x-ray imaging system according to some embodiments of inventive concepts;

[0051] FIG. **32** is a flow chart illustrating use of a calibrated fixture according to some embodiments of inventive concepts;

[0052] FIG. **33** is a schematic diagram illustrating a fixture for use on an x-ray detector according to some embodiments of inventive concepts;

[0053] FIG. **34** is a block diagram illustrating a fixture controller according to some embodiments of inventive concepts;

[0054] FIG. **35** is a block diagram illustrating a robotic controller according to some embodiments of inventive concepts;

[0055] FIGS. **36A and 36B** are flow charts illustrating operations of a fixture according to some embodiments of inventive concepts;

[0056] FIG. **37** is a flow chart illustrating operations of a robotic system according to some embodiments of inventive concepts;

[0057] FIGS. **38A and 38B** are plan views illustrating a fixture of FIG. **33** according to some embodiments of inventive concepts;

[0058] FIG. **39** is a plan view of a fluoroscopic x-ray imaging system including mounting sockets for a fixture according to some embodiments of inventive concepts;

[0059] FIG. **40** is a plan view of a fixture mounted on an x-ray detector of a fluoroscopic x-ray imaging system according to some embodiments of inventive concepts;

[0060] FIG. **41** is a plan view of a fixture and a temporary x-ray opaque fiducial pattern mounted on an x-ray detector of a fluoroscopic x-ray imaging system according to some embodiments of inventive concepts;

[0061] FIG. **42** is an expanded plan view of an x-ray detector of a fluoroscopic x-ray imaging

system including mounting sockets according to some embodiments of inventive concepts; [0062] FIG. **43** is a an expanded plan view of a fixture mounted on an x-ray detector of a fluoroscopic x-ray imaging system according to some embodiments of inventive concepts; [0063] FIG. **44** is an expanded plan view of a fixture and a temporary x-ray opaque fiducial pattern mounted on an x-ray detector of a fluoroscopic x-ray imaging system according to some embodiments of inventive concepts; [0064] FIGS. **45A**, **45B**, and **45C** are enlarged plan views illustrating a fixture mounted on an x-ray detector of a fluoroscopic imaging system, a sterile drape covering the fixture and the x-ray detector, and application of tracking markers over/through the sterile drape according to some embodiments of inventive concepts.

DETAILED DESCRIPTION

[0065] It is to be understood that the present disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the description herein or illustrated in the drawings. The teachings of the present disclosure may be used and practiced in other embodiments and practiced or carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings. Further, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings.

[0066] The following discussion is presented to enable a person skilled in the art to make and use embodiments of the present disclosure. Various modifications to the illustrated embodiments will be readily apparent to those skilled in the art, and the principles herein can be applied to other embodiments and applications without departing from embodiments of the present disclosure. Thus, the embodiments are not intended to be limited to embodiments shown, but are to be accorded the widest scope consistent with the principles and features disclosed herein. The following detailed description is to be read with reference to the figures, in which like elements in different figures have like reference numerals. The figures, which are not necessarily to scale, depict selected embodiments and are not intended to limit the scope of the embodiments. Skilled artisans will recognize the examples provided herein have many useful alternatives and fall within the scope of the embodiments.

[0067] Turning now to the drawing, FIGS. **1** and **2** illustrate a surgical robot system **100** in accordance with an exemplary embodiment. Surgical robot system **100** may include, for example, a surgical robot **102**, one or more robot arms **104**, a base **106**, a display **110**, an end-effector **112**, for example, including a guide tube **114**, and one or more tracking markers **118**. The surgical robot system **100** may include a patient tracking device **116** also including one or more tracking markers **118**, which is adapted to be secured directly to the patient **210** (e.g., to a bone of the patient **210**). The surgical robot system **100** may also use a camera **200**, for example, positioned on a camera stand **202**. The camera stand **202** can have any suitable configuration to move, orient, and support the camera **200** in a desired position. The camera **200** may include any suitable camera or cameras, such as one or more infrared cameras (e.g., bifocal or stereophotogrammetric cameras), able to identify, for example, active and passive tracking markers **118** (shown as part of patient tracking device **116** in FIG. **2** and shown by enlarged view in FIGS. **13A-13B**) in a given measurement volume viewable from the perspective of the camera **200**. The camera **200** may scan the given measurement volume and detect the light that comes from the markers **118** in order to identify and determine the position of the markers **118** in three-dimensions. For example, active markers **118** may include infrared-emitting markers that are activated by an electrical signal (e.g., infrared light emitting diodes (LEDs)), and/or passive markers **118** may include retro-reflective markers that

reflect infrared light (e.g., they reflect incoming IR radiation into the direction of the incoming light), for example, emitted by illuminators on the camera **200** or other suitable device.

[0068] FIGS. **1** and **2** illustrate a potential configuration for the placement of the surgical robot system **100** in an operating room environment. For example, the robot **102** may be positioned near or next to patient **210**. Although depicted near the head of the patient **210**, it will be appreciated that the robot **102** can be positioned at any suitable location near the patient **210** depending on the area of the patient **210** undergoing the operation. The camera **200** may be separated from the robot system **100** and positioned at the foot of patient **210**. This location allows the camera **200** to have a direct visual line of sight to the surgical field **208**. Again, it is contemplated that the camera **200** may be located at any suitable position having line of sight to the surgical field **208**. In the configuration shown, the surgeon **120** may be positioned across from the robot **102**, but is still able to manipulate the end-effector **112** and the display **110**. A surgical assistant **126** may be positioned across from the surgeon **120** again with access to both the end-effector **112** and the display **110**. If desired, the locations of the surgeon **120** and the assistant **126** may be reversed. The traditional areas for the anesthesiologist **122** and the nurse or scrub tech **124** may remain unimpeded by the locations of the robot **102** and camera **200**.

[0069] With respect to the other components of the robot **102**, the display **110** can be attached to the surgical robot **102** and in other exemplary embodiments, display **110** can be detached from surgical robot **102**, either within a surgical room with the surgical robot **102**, or in a remote location. End-effector **112** may be coupled to the robot arm **104** and controlled by at least one motor. In exemplary embodiments, end-effector **112** can comprise a guide tube **114**, which is able to receive and orient a surgical instrument **608** (described further herein) used to perform surgery on the patient **210**. As used herein, the term “end-effector” is used interchangeably with the terms “end-effectuator” and “effectuator element.” Although generally shown with a guide tube **114**, it will be appreciated that the end-effector **112** may be replaced with any suitable instrumentation suitable for use in surgery. In some embodiments, end-effector **112** can comprise any known structure for effecting the movement of the surgical instrument **608** in a desired manner.

[0070] The surgical robot **102** is able to control the translation and orientation of the end-effector **112**. The robot **102** is able to move end-effector **112** along x-, y-, and z-axes, for example. The end-effector **112** can be configured for selective rotation about one or more of the x-, y-, and z-axis, and a Z Frame axis (such that one or more of the Euler Angles (e.g., roll, pitch, and/or yaw) associated with end-effector **112** can be selectively controlled). In some exemplary embodiments, selective control of the translation and orientation of end-effector **112** can permit performance of medical procedures with significantly improved accuracy compared to conventional robots that use, for example, a six degree of freedom robot arm comprising only rotational axes. For example, the surgical robot system **100** may be used to operate on patient **210**, and robot arm **104** can be positioned above the body of patient **210**, with end-effector **112** selectively angled relative to the z-axis toward the body of patient **210**.

[0071] In some exemplary embodiments, the position of the surgical instrument **608** can be dynamically updated so that surgical robot **102** can be aware of the location of the surgical instrument **608** at all times during the procedure. Consequently, in some exemplary embodiments, surgical robot **102** can move the surgical instrument **608** to the desired position quickly without any further assistance from a physician (unless the physician so desires). In some further embodiments, surgical robot **102** can be configured to correct the path of the surgical instrument **608** if the surgical instrument **608** strays from the selected, preplanned trajectory. In some exemplary embodiments, surgical robot **102** can be configured to permit stoppage, modification, and/or manual control of the movement of end-effector **112** and/or the surgical instrument **608**. Thus, in use, in exemplary embodiments, a physician or other user can operate the system **100**, and has the option to stop, modify, or manually control the autonomous movement of end-effector **112** and/or the surgical instrument **608**. Further details of surgical robot system **100** including the control and

movement of a surgical instrument **608** by surgical robot **102** can be found in co-pending U.S. Pat. No. 9,782,229, the disclosure of which is hereby incorporated herein by reference in its entirety. [0072] The robotic surgical system **100** can comprise one or more tracking markers **118** configured to track the movement of robot arm **104**, end-effector **112**, patient **210**, and/or the surgical instrument **608** in three dimensions. In exemplary embodiments, a plurality of tracking markers **118** can be mounted (or otherwise secured) thereon to an outer surface of the robot **102**, such as, for example and without limitation, on base **106** of robot **102**, on robot arm **104**, and/or on the end-effector **112**. In exemplary embodiments, at least one tracking marker **118** of the plurality of tracking markers **118** can be mounted or otherwise secured to the end-effector **112**. One or more tracking markers **118** can further be mounted (or otherwise secured) to the patient **210**. In exemplary embodiments, the plurality of tracking markers **118** can be positioned on the patient **210** spaced apart from the surgical field **208** to reduce the likelihood of being obscured by the surgeon, surgical tools, or other parts of the robot **102**. Further, one or more tracking markers **118** can be further mounted (or otherwise secured) to the surgical tools **608** (e.g., a screwdriver, dilator, implant inserter, or the like). Thus, the tracking markers **118** enable each of the marked objects (e.g., the end-effector **112**, the patient **210**, and the surgical tools **608**) to be tracked by the robot **102**. In exemplary embodiments, system **100** can use tracking information collected from each of the marked objects to calculate the orientation and location, for example, of the end-effector **112**, the surgical instrument **608** (e.g., positioned in the tube **114** of the end-effector **112**), and the relative position of the patient **210**.

[0073] The markers **118** may include radiopaque or optical markers. The markers **118** may be suitably shaped include spherical, spheroid, cylindrical, cube, cuboid, or the like. In exemplary embodiments, one or more of markers **118** may be optical markers. In some embodiments, the positioning of one or more tracking markers **118** on end-effector **112** may increase/maximize accuracy of positional measurements by serving to check or verify a position of end-effector **112**. Further details of surgical robot system **100** including the control, movement and tracking of surgical robot **102** and of a surgical instrument **608** can be found in U.S. patent publication No. 2016/0242849, the disclosure of which is incorporated herein by reference in its entirety.

[0074] Exemplary embodiments include one or more markers **118** coupled to the surgical instrument **608**. In exemplary embodiments, these markers **118**, for example, coupled to the patient **210** and surgical instruments **608**, as well as markers **118** coupled to the end-effector **112** of the robot **102** can comprise conventional infrared light-emitting diodes (LEDs) or an Optotrak® diode capable of being tracked using a commercially available infrared optical tracking system such as Optotrak®. Optotrak® is a registered trademark of Northern Digital Inc., Waterloo, Ontario, Canada. In other embodiments, markers **118** can comprise conventional reflective spheres capable of being tracked using a commercially available optical tracking system such as Polaris Spectra. Polaris Spectra is also a registered trademark of Northern Digital, Inc. In an exemplary embodiment, the markers **118** coupled to the end-effector **112** are active markers which comprise infrared light-emitting diodes which may be turned on and off, and the markers **118** coupled to the patient **210** and the surgical instruments **608** comprise passive reflective spheres.

[0075] In exemplary embodiments, light emitted from and/or reflected by markers **118** can be detected by camera **200** and can be used to monitor the location and movement of the marked objects. In alternative embodiments, markers **118** can comprise a radio-frequency and/or electromagnetic reflector or transceiver and the camera **200** can include or be replaced by a radio-frequency and/or electromagnetic transceiver.

[0076] Similar to surgical robot system **100**, FIG. 3 illustrates a surgical robot system **300** and camera stand **302**, in a docked configuration, consistent with an exemplary embodiment of the present disclosure. Surgical robot system **300** may comprise a robot **301** including a display **304**, upper arm **306**, lower arm **308**, end-effector **310**, vertical column **312**, casters **314**, cabinet **316**, tablet drawer **318**, connector panel **320**, control panel **322**, and ring of information **324**. Camera

stand **302** may comprise camera **326**. These components are described in greater with respect to FIG. 5. FIG. 3 illustrates the surgical robot system **300** in a docked configuration where the camera stand **302** is nested with the robot **301**, for example, when not in use. It will be appreciated by those skilled in the art that the camera **326** and robot **301** may be separated from one another and positioned at any appropriate location during the surgical procedure, for example, as shown in FIGS. 1 and 2.

[0077] FIG. 4 illustrates a base **400** consistent with an exemplary embodiment of the present disclosure. Base **400** may be a portion of surgical robot system **300** and comprise cabinet **316**. Cabinet **316** may house certain components of surgical robot system **300** including but not limited to a battery **402**, a power distribution module **404**, a platform interface board module **406**, a computer **408**, a handle **412**, and a tablet drawer **414**. The connections and relationship between these components is described in greater detail with respect to FIG. 5.

[0078] FIG. 5 illustrates a block diagram of certain components of an exemplary embodiment of surgical robot system **300**. Surgical robot system **300** may comprise platform subsystem **502**, computer subsystem **504**, motion control subsystem **506**, and tracking subsystem **532**. Platform subsystem **502** may further comprise battery **402**, power distribution module **404**, platform interface board module **406**, and tablet charging station **534**. Computer subsystem **504** may further comprise computer **408**, display **304**, and speaker **536**. Motion control subsystem **506** may further comprise driver circuit **508**, motors **510**, **512**, **514**, **516**, **518**, stabilizers **520**, **522**, **524**, **526**, end-effector **310**, and controller **538**. Tracking subsystem **532** may further comprise position sensor **540** and camera converter **542**. System **300** may also comprise a foot pedal **544** and tablet **546**.

[0079] Input power is supplied to system **300** via a power source **548** which may be provided to power distribution module **404**. Power distribution module **404** receives input power and is configured to generate different power supply voltages that are provided to other modules, components, and subsystems of system **300**. Power distribution module **404** may be configured to provide different voltage supplies to platform interface module **406**, which may be provided to other components such as computer **408**, display **304**, speaker **536**, driver **508** to, for example, power motors **512**, **514**, **516**, **518** and end-effector **310**, motor **510**, ring **324**, camera converter **542**, and other components for system **300** for example, fans for cooling the electrical components within cabinet **316**.

[0080] Power distribution module **404** may also provide power to other components such as tablet charging station **534** that may be located within tablet drawer **318**. Tablet charging station **534** may be in wireless or wired communication with tablet **546** for charging table **546**. Tablet **546** may be used by a surgeon consistent with the present disclosure and described herein.

[0081] Power distribution module **404** may also be connected to battery **402**, which serves as temporary power source in the event that power distribution module **404** does not receive power from input power **548**. At other times, power distribution module **404** may serve to charge battery **402** if necessary.

[0082] Other components of platform subsystem **502** may also include connector panel **320**, control panel **322**, and ring **324**. Connector panel **320** may serve to connect different devices and components to system **300** and/or associated components and modules. Connector panel **320** may contain one or more ports that receive lines or connections from different components. For example, connector panel **320** may have a ground terminal port that may ground system **300** to other equipment, a port to connect foot pedal **544** to system **300**, a port to connect to tracking subsystem **532**, which may comprise position sensor **540**, camera converter **542**, and cameras **326** associated with camera stand **302**. Connector panel **320** may also include other ports to allow USB, Ethernet, HDMI communications to other components, such as computer **408**.

[0083] Control panel **322** may provide various buttons or indicators that control operation of system **300** and/or provide information regarding system **300**. For example, control panel **322** may include buttons to power on or off system **300**, lift or lower vertical column **312**, and lift or lower

stabilizers **520-526** that may be designed to engage casters **314** to lock system **300** from physically moving. Other buttons may stop system **300** in the event of an emergency, which may remove all motor power and apply mechanical brakes to stop all motion from occurring. Control panel **322** may also have indicators notifying the user of certain system conditions such as a line power indicator or status of charge for battery **402**.

[0084] Ring **324** may be a visual indicator to notify the user of system **300** of different modes that system **300** is operating under and certain warnings to the user.

[0085] Computer subsystem **504** includes computer **408**, display **304**, and speaker **536**. Computer **504** includes an operating system and software to operate system **300**. Computer **504** may receive and process information from other components (for example, tracking subsystem **532**, platform subsystem **502**, and/or motion control subsystem **506**) in order to display information to the user. Further, computer subsystem **504** may also include speaker **536** to provide audio to the user.

[0086] Tracking subsystem **532** may include position sensor **504** and converter **542**. Tracking subsystem **532** may correspond to camera stand **302** including camera **326** as described with respect to FIG. 3. Position sensor **504** may be camera **326**. Tracking subsystem may track the location of certain markers that are located on the different components of system **300** and/or instruments used by a user during a surgical procedure. This tracking may be conducted in a manner consistent with the present disclosure including the use of infrared technology that tracks the location of active or passive elements, such as LEDs or reflective markers, respectively. The location, orientation, and position of structures having these types of markers may be provided to computer **408** which may be shown to a user on display **304**. For example, a surgical instrument **608** having these types of markers and tracked in this manner (which may be referred to as a navigational space) may be shown to a user in relation to a three dimensional image of a patient's anatomical structure.

[0087] Motion control subsystem **506** may be configured to physically move vertical column **312**, upper arm **306**, lower arm **308**, or rotate end-effector **310**. The physical movement may be conducted through the use of one or more motors **510-518**. For example, motor **510** may be configured to vertically lift or lower vertical column **312**. Motor **512** may be configured to laterally move upper arm **308** around a point of engagement with vertical column **312** as shown in FIG. 3. Motor **514** may be configured to laterally move lower arm **308** around a point of engagement with upper arm **308** as shown in FIG. 3. Motors **516** and **518** may be configured to move end-effector **310** in a manner such that one may control the roll and one may control the tilt, thereby providing multiple angles that end-effector **310** may be moved. These movements may be achieved by controller **538** which may control these movements through load cells disposed on end-effector **310** and activated by a user engaging these load cells to move system **300** in a desired manner.

[0088] Moreover, system **300** may provide for automatic movement of vertical column **312**, upper arm **306**, and lower arm **308** through a user indicating on display **304** (which may be a touchscreen input device) the location of a surgical instrument or component on a three dimensional image of the patient's anatomy on display **304**. The user may initiate this automatic movement by stepping on foot pedal **544** or some other input means.

[0089] FIG. 6 illustrates a surgical robot system **600** consistent with an exemplary embodiment. Surgical robot system **600** may comprise end-effector **602**, robot arm **604**, guide tube **606**, instrument **608**, and robot base **610**. Instrument tool **608** may be attached to a tracking array **612** including one or more tracking markers (such as markers **118**) and have an associated trajectory **614**. Trajectory **614** may represent a path of movement that instrument tool **608** is configured to travel once it is positioned through or secured in guide tube **606**, for example, a path of insertion of instrument tool **608** into a patient. In an exemplary operation, robot base **610** may be configured to be in electronic communication with robot arm **604** and end-effector **602** so that surgical robot system **600** may assist a user (for example, a surgeon) in operating on the patient **210**. Surgical robot system **600** may be consistent with previously described surgical robot system **100** and **300**.

[0090] A tracking array **612** may be mounted on instrument **608** to monitor the location and orientation of instrument tool **608**. The tracking array **612** may be attached to an instrument **608** and may comprise tracking markers **804**. As best seen in FIG. **8**, tracking markers **804** may be, for example, light emitting diodes and/or other types of reflective markers (e.g., markers **118** as described elsewhere herein). The tracking devices may be one or more line of sight devices associated with the surgical robot system. As an example, the tracking devices may be one or more cameras **200**, **326** associated with the surgical robot system **100**, **300** and may also track tracking array **612** for a defined domain or relative orientations of the instrument **608** in relation to the robot arm **604**, the robot base **610**, end-effector **602**, and/or the patient **210**. The tracking devices may be consistent with those structures described in connection with camera stand **302** and tracking subsystem **532**.

[0091] FIGS. **7A**, **7B**, and **7C** illustrate a top view, front view, and side view, respectively, of end-effector **602** consistent with an exemplary embodiment. End-effector **602** may comprise one or more tracking markers **702**. Tracking markers **702** may be light emitting diodes or other types of active and passive markers, such as tracking markers **118** that have been previously described. In an exemplary embodiment, the tracking markers **702** are active infrared-emitting markers that are activated by an electrical signal (e.g., infrared light emitting diodes (LEDs)). Thus, tracking markers **702** may be activated such that the infrared markers **702** are visible to the camera **200**, **326** or may be deactivated such that the infrared markers **702** are not visible to the camera **200**, **326**. Thus, when the markers **702** are active, the end-effector **602** may be controlled by the system **100**, **300**, **600**, and when the markers **702** are deactivated, the end-effector **602** may be locked in position and unable to be moved by the system **100**, **300**, **600**.

[0092] Markers **702** may be disposed on or within end-effector **602** in a manner such that the markers **702** are visible by one or more cameras **200**, **326** or other tracking devices associated with the surgical robot system **100**, **300**, **600**. The camera **200**, **326** or other tracking devices may track end-effector **602** as it moves to different positions and viewing angles by following the movement of tracking markers **702**. The location of markers **702** and/or end-effector **602** may be shown on a display **110**, **304** associated with the surgical robot system **100**, **300**, **600**, for example, display **110** as shown in FIG. **2** and/or display **304** shown in FIG. **3**. This display **110**, **304** may allow a user to ensure that end-effector **602** is in a desirable position in relation to robot arm **604**, robot base **610**, the patient **210**, and/or the user.

[0093] For example, as shown in FIG. **7A**, markers **702** may be placed around the surface of end-effector **602** so that a tracking device placed away from the surgical field **208** and facing toward the robot **102**, **301** and the camera **200**, **326** is able to view at least 3 of the markers **702** through a range of common orientations of the end-effector **602** relative to the tracking device. For example, distribution of markers **702** in this way allows end-effector **602** to be monitored by the tracking devices when end-effector **602** is translated and rotated in the surgical field **208**.

[0094] In addition, in exemplary embodiments, end-effector **602** may be equipped with infrared (IR) receivers that can detect when an external camera **200**, **326** is getting ready to read markers **702**. Upon this detection, end-effector **602** may then illuminate markers **702**. The detection by the IR receivers that the external camera **200**, **326** is ready to read markers **702** may signal the need to synchronize a duty cycle of markers **702**, which may be light emitting diodes, to an external camera **200**, **326**. This may also allow for lower power consumption by the robotic system as a whole, whereby markers **702** would only be illuminated at the appropriate time instead of being illuminated continuously. Further, in exemplary embodiments, markers **702** may be powered off to prevent interference with other navigation tools, such as different types of surgical instruments **608**.

[0095] FIG. **8** depicts one type of surgical instrument **608** including a tracking array **612** and tracking markers **804**. Tracking markers **804** may be of any type described herein including but not limited to light emitting diodes or reflective spheres. Markers **804** are monitored by tracking devices associated with the surgical robot system **100**, **300**, **600** and may be one or more of the line

of sight cameras **200, 326**. The cameras **200, 326** may track the location of instrument **608** based on the position and orientation of tracking array **612** and markers **804**. A user, such as a surgeon **120**, may orient instrument **608** in a manner so that tracking array **612** and markers **804** are sufficiently recognized by the tracking device or camera **200, 326** to display instrument **608** and markers **804** on, for example, display **110** of the exemplary surgical robot system.

[0096] The manner in which a surgeon **120** may place instrument **608** into guide tube **606** of the end-effector **602** and adjust the instrument **608** is evident in FIG. **8**. The hollow tube or guide tube **114, 606** of the end-effector **112, 310, 602** is sized and configured to receive at least a portion of the surgical instrument **608**. The guide tube **114, 606** is configured to be oriented by the robot arm **104** such that insertion and trajectory for the surgical instrument **608** is able to reach a desired anatomical target within or upon the body of the patient **210**. The surgical instrument **608** may include at least a portion of a generally cylindrical instrument. Although a screwdriver is exemplified as the surgical tool **608**, it will be appreciated that any suitable surgical tool **608** may be positioned by the end-effector **602**. By way of example, the surgical instrument **608** may include one or more of a guide wire, cannula, a retractor, a drill, a reamer, a screwdriver, an insertion tool, a removal tool, or the like. Although the hollow tube **114, 606** is generally shown as having a cylindrical configuration, it will be appreciated by those of skill in the art that the guide tube **114, 606** may have any suitable shape, size and configuration desired to accommodate the surgical instrument **608** and access the surgical site.

[0097] FIGS. **9A-9C** illustrate end-effector **602** and a portion of robot arm **604** consistent with an exemplary embodiment. End-effector **602** may further comprise body **1202** and clamp **1204**. Clamp **1204** may comprise handle **1206**, balls **1208**, spring **1210**, and lip **1212**. Robot arm **604** may further comprise depressions **1214**, mounting plate **1216**, lip **1218**, and magnets **1220**.

[0098] End-effector **602** may mechanically interface and/or engage with the surgical robot system and robot arm **604** through one or more couplings. For example, end-effector **602** may engage with robot arm **604** through a locating coupling and/or a reinforcing coupling. Through these couplings, end-effector **602** may fasten with robot arm **604** outside a flexible and sterile barrier. In an exemplary embodiment, the locating coupling may be a magnetically kinematic mount and the reinforcing coupling may be a five bar over center clamping linkage.

[0099] With respect to the locating coupling, robot arm **604** may comprise mounting plate **1216**, which may be non-magnetic material, one or more depressions **1214**, lip **1218**, and magnets **1220**. Magnet **1220** is mounted below each of depressions **1214**. Portions of clamp **1204** may comprise magnetic material and be attracted by one or more magnets **1220**. Through the magnetic attraction of clamp **1204** and robot arm **604**, balls **1208** become seated into respective depressions **1214**. For example, balls **1208** as shown in FIG. **9B** would be seated in depressions **1214** as shown in FIG. **9A**. This seating may be considered a magnetically-assisted kinematic coupling. Magnets **1220** may be configured to be strong enough to support the entire weight of end-effector **602** regardless of the orientation of end-effector **602**. The locating coupling may be any style of kinematic mount that uniquely restrains six degrees of freedom.

[0100] With respect to the reinforcing coupling, portions of clamp **1204** may be configured to be a fixed ground link and as such clamp **1204** may serve as a five bar linkage. Closing clamp handle **1206** may fasten end-effector **602** to robot arm **604** as lip **1212** and lip **1218** engage clamp **1204** in a manner to secure end-effector **602** and robot arm **604**. When clamp handle **1206** is closed, spring **1210** may be stretched or stressed while clamp **1204** is in a locked position. The locked position may be a position that provides for linkage past center. Because of a closed position that is past center, the linkage will not open absent a force applied to clamp handle **1206** to release clamp **1204**. Thus, in a locked position, end-effector **602** may be robustly secured to robot arm **604**.

[0101] Spring **1210** may be a curved beam in tension. Spring **1210** may be comprised of a material that exhibits high stiffness and high yield strain such as virgin PEEK (poly-ether-ether-ketone). The linkage between end-effector **602** and robot arm **604** may provide for a sterile barrier between end-

effector **602** and robot arm **604** without impeding fastening of the two couplings.

[0102] The reinforcing coupling may be a linkage with multiple spring members. The reinforcing coupling may latch with a cam or friction based mechanism. The reinforcing coupling may also be a sufficiently powerful electromagnet that will support fastening end-effector **102** to robot arm **604**. The reinforcing coupling may be a multi-piece collar completely separate from either end-effector **602** and/or robot arm **604** that slips over an interface between end-effector **602** and robot arm **604** and tightens with a screw mechanism, an over center linkage, or a cam mechanism.

[0103] Referring to FIGS. **10** and **11**, prior to or during a surgical procedure, certain registration procedures may be conducted to track objects and a target anatomical structure of the patient **210** both in a navigation space and an image space. To conduct such registration, a registration system **1400** may be used as illustrated in FIG. **10**.

[0104] To track the position of the patient **210**, a patient tracking device **116** may include a patient fixation instrument **1402** to be secured to a rigid anatomical structure of the patient **210** and a dynamic reference base (DRB) **1404** may be securely attached to the patient fixation instrument **1402**. For example, patient fixation instrument **1402** may be inserted into opening **1406** of dynamic reference base **1404**. Dynamic reference base **1404** may contain markers **1408** that are visible to tracking devices, such as tracking subsystem **532**. These markers **1408** may be optical markers or reflective spheres, such as tracking markers **118**, as previously discussed herein.

[0105] Patient fixation instrument **1402** is attached to a rigid anatomy of the patient **210** and may remain attached throughout the surgical procedure. In an exemplary embodiment, patient fixation instrument **1402** is attached to a rigid area of the patient **210**, for example, a bone that is located away from the targeted anatomical structure subject to the surgical procedure. In order to track the targeted anatomical structure, dynamic reference base **1404** is associated with the targeted anatomical structure through the use of a registration fixture that is temporarily placed on or near the targeted anatomical structure in order to register the dynamic reference base **1404** with the location of the targeted anatomical structure.

[0106] A registration fixture **1410** is attached to patient fixation instrument **1402** through the use of a pivot arm **1412**. Pivot arm **1412** is attached to patient fixation instrument **1402** by inserting patient fixation instrument **1402** through an opening **1414** of registration fixture **1410**. Pivot arm **1412** is attached to registration fixture **1410** by, for example, inserting a knob **1416** through an opening **1418** of pivot arm **1412**.

[0107] Using pivot arm **1412**, registration fixture **1410** may be placed over the targeted anatomical structure and its location may be determined in an image space and navigation space using tracking markers **1420** and/or fiducials **1422** on registration fixture **1410**. Registration fixture **1410** may contain a collection of markers **1420** that are visible in a navigational space (for example, markers **1420** may be detectable by tracking subsystem **532**). Tracking markers **1420** may be optical markers visible in infrared light as previously described herein. Registration fixture **1410** may also contain a collection of fiducials **1422**, for example, such as bearing balls, that are visible in an imaging space (for example, a three dimension CT image). As described in greater detail with respect to FIG. **11**, using registration fixture **1410**, the targeted anatomical structure may be associated with dynamic reference base **1404** thereby allowing depictions of objects in the navigational space to be overlaid on images of the anatomical structure. Dynamic reference base **1404**, located at a position away from the targeted anatomical structure, may become a reference point thereby allowing removal of registration fixture **1410** and/or pivot arm **1412** from the surgical area.

[0108] FIG. **11** provides an exemplary method **1500** for registration consistent with the present disclosure. Method **1500** begins at step **1502** wherein a graphical representation (or image(s)) of the targeted anatomical structure may be imported into system **100**, **300** **600**, for example computer **408**. The graphical representation may be three dimensional CT or a fluoroscope scan of the targeted anatomical structure of the patient **210** which includes registration fixture **1410** and a

detectable imaging pattern of fiducials **1420**.

[0109] At step **1504**, an imaging pattern of fiducials **1420** is detected and registered in the imaging space and stored in computer **408**. Optionally, at this time at step **1506**, a graphical representation of the registration fixture **1410** may be overlaid on the images of the targeted anatomical structure.

[0110] At step **1508**, a navigational pattern of registration fixture **1410** is detected and registered by recognizing markers **1420**. Markers **1420** may be optical markers that are recognized in the navigation space through infrared light by tracking subsystem **532** via position sensor **540**. Thus, the location, orientation, and other information of the targeted anatomical structure is registered in the navigation space. Therefore, registration fixture **1410** may be recognized in both the image space through the use of fiducials **1422** and the navigation space through the use of markers **1420**. At step **1510**, the registration of registration fixture **1410** in the image space is transferred to the navigation space. This transferal is done, for example, by using the relative position of the imaging pattern of fiducials **1422** compared to the position of the navigation pattern of markers **1420**.

[0111] At step **1512**, registration of the navigation space of registration fixture **1410** (having been registered with the image space) is further transferred to the navigation space of dynamic registration array **1404** attached to patient fixture instrument **1402**. Thus, registration fixture **1410** may be removed and dynamic reference base **1404** may be used to track the targeted anatomical structure in both the navigation and image space because the navigation space is associated with the image space.

[0112] At steps **1514** and **1516**, the navigation space may be overlaid on the image space and objects with markers visible in the navigation space (for example, surgical instruments **608** with optical markers **804**). The objects may be tracked through graphical representations of the surgical instrument **608** on the images of the targeted anatomical structure.

[0113] FIGS. **12A-12B** illustrate imaging devices **1304** that may be used in conjunction with robot systems **100**, **300**, **600** to acquire pre-operative, intra-operative, post-operative, and/or real-time image data of patient **210**. Any appropriate subject matter may be imaged for any appropriate procedure using the imaging system **1304**. The imaging system **1304** may be any imaging device such as imaging device **1306** and/or a C-arm **1308** device. It may be desirable to take x-rays of patient **210** from a number of different positions, without the need for frequent manual repositioning of patient **210** which may be required in an x-ray system. As illustrated in FIG. **12A**, the imaging system **1304** may be in the form of a C-arm **1308** that includes an elongated C-shaped member terminating in opposing distal ends **1312** of the “C” shape. C-shaped member **1130** may further comprise an x-ray source **1314** and an x-ray detector **1316** (also referred to as an image receptor). The space within C-arm **1308** of the arm may provide room for the physician to attend to the patient substantially free of interference from x-ray support structure **1318**. As illustrated in FIG. **12B**, the imaging system may include imaging device **1306** having a gantry housing **1324** attached to a support structure imaging device support structure **1328**, such as a wheeled mobile cart **1330** with wheels **1332**, which may enclose an image capturing portion, not illustrated. The image capturing portion may include an x-ray source and/or emission portion and an x-ray receiving and/or image receiving portion, which may be disposed about one hundred and eighty degrees from each other and mounted on a rotor (not illustrated) relative to a track of the image capturing portion. The image capturing portion may be operable to rotate three hundred and sixty degrees during image acquisition. The image capturing portion may rotate around a central point and/or axis, allowing image data of patient **210** to be acquired from multiple directions or in multiple planes. Although certain imaging systems **1304** are exemplified herein, it will be appreciated that any suitable imaging system may be selected by one of ordinary skill in the art.

[0114] Turning now to FIGS. **13A-13C**, the surgical robot system **100**, **300**, **600** relies on accurate positioning of the end-effector **112**, **602**, surgical instruments **608**, and/or the patient **210** (e.g., patient tracking device **116**) relative to the desired surgical area. In the embodiments shown in FIGS. **13A-13C**, the tracking markers **118**, **804** are rigidly attached to a portion of the instrument

608 and/or end-effector **112**.

[0115] FIG. **13A** depicts part of the surgical robot system **100** with the robot **102** including base **106**, robot arm **104**, and end-effector **112**. The other elements, not illustrated, such as the display, cameras, etc. may also be present as described herein. FIG. **13B** depicts a close-up view of the end-effector **112** with guide tube **114** and a plurality of tracking markers **118** rigidly affixed to the end-effector **112**. In this embodiment, the plurality of tracking markers **118** are attached to the guide tube **112**. FIG. **13C** depicts an instrument **608** (in this case, a probe **608A**) with a plurality of tracking markers **804** rigidly affixed to the instrument **608**. As described elsewhere herein, the instrument **608** could include any suitable surgical instrument, such as, but not limited to, guide wire, cannula, a retractor, a drill, a reamer, a screwdriver, an insertion tool, a removal tool, or the like.

[0116] When tracking an instrument **608**, end-effector **112**, or other object to be tracked in 3D, an array of tracking markers **118**, **804** may be rigidly attached to a portion of the tool **608** or end-effector **112**. Preferably, the tracking markers **118**, **804** are attached such that the markers **118**, **804** are out of the way (e.g., not impeding the surgical operation, visibility, etc.). The markers **118**, **804** may be affixed to the instrument **608**, end-effector **112**, or other object to be tracked, for example, with an array **612**. Usually three or four markers **118**, **804** are used with an array **612**. The array **612** may include a linear section, a cross piece, and may be asymmetric such that the markers **118**, **804** are at different relative positions and locations with respect to one another. For example, as shown in FIG. **13C**, a probe **608A** with a 4-marker tracking array **612** is shown, and FIG. **13B** depicts the end-effector **112** with a different 4-marker tracking array **612**.

[0117] In FIG. **13C**, the tracking array **612** functions as the handle **620** of the probe **608A**. Thus, the four markers **804** are attached to the handle **620** of the probe **608A**, which is out of the way of the shaft **622** and tip **624**. Stereophotogrammetric tracking of these four markers **804** allows the instrument **608** to be tracked as a rigid body and for the tracking system **100**, **300**, **600** to precisely determine the position of the tip **624** and the orientation of the shaft **622** while the probe **608A** is moved around in front of tracking cameras **200**, **326**.

[0118] To enable automatic tracking of one or more tools **608**, end-effector **112**, or other object to be tracked in 3D (e.g., multiple rigid bodies), the markers **118**, **804** on each tool **608**, end-effector **112**, or the like, are arranged asymmetrically with a known inter-marker spacing. The reason for asymmetric alignment is so that it is unambiguous which marker **118**, **804** corresponds to a particular location on the rigid body and whether markers **118**, **804** are being viewed from the front or back, i.e., mirrored. For example, if the markers **118**, **804** were arranged in a square on the tool **608** or end-effector **112**, it would be unclear to the system **100**, **300**, **600** which marker **118**, **804** corresponded to which corner of the square. For example, for the probe **608A**, it would be unclear which marker **804** was closest to the shaft **622**. Thus, it would be unknown which way the shaft **622** was extending from the array **612**. Accordingly, each array **612** and thus each tool **608**, end-effector **112**, or other object to be tracked should have a unique marker pattern to allow it to be distinguished from other tools **608** or other objects being tracked. Asymmetry and unique marker patterns allow the system **100**, **300**, **600** to detect individual markers **118**, **804** then to check the marker spacing against a stored template to determine which tool **608**, end effector **112**, or other object they represent. Detected markers **118**, **804** can then be sorted automatically and assigned to each tracked object in the correct order. Without this information, rigid body calculations could not then be performed to extract key geometric information, for example, such as tool tip **624** and alignment of the shaft **622**, unless the user manually specified which detected marker **118**, **804** corresponded to which position on each rigid body. These concepts are commonly known to those skilled in the methods of 3D optical tracking.

[0119] Turning now to FIGS. **14A-14D**, an alternative version of an end-effector **912** with moveable tracking markers **918A-918D** is shown. In FIG. **14A**, an array with moveable tracking markers **918A-918D** are shown in a first configuration, and in FIG. **14B** the moveable tracking

markers **918A-918D** are shown in a second configuration, which is angled relative to the first configuration. FIG. **14C** shows the template of the tracking markers **918A-918D**, for example, as seen by the cameras **200, 326** in the first configuration of FIG. **14A**; and FIG. **14D** shows the template of tracking markers **918A-918D**, for example, as seen by the cameras **200, 326** in the second configuration of FIG. **14B**.

[0120] In this embodiment, 4-marker array tracking is contemplated wherein the markers **918A-918D** are not all in fixed position relative to the rigid body and instead, one or more of the array markers **918A-918D** can be adjusted, for example, during testing, to give updated information about the rigid body that is being tracked without disrupting the process for automatic detection and sorting of the tracked markers **918A-918D**.

[0121] When tracking any tool, such as a guide tube **914** connected to the end effector **912** of a robot system **100, 300, 600**, the tracking array's primary purpose is to update the position of the end effector **912** in the camera coordinate system. When using the rigid system, for example, as shown in FIG. **13B**, the array **612** of reflective markers **118** rigidly extend from the guide tube **114**.

Because the tracking markers **118** are rigidly connected, knowledge of the marker locations in the camera coordinate system also provides exact location of the centerline, tip, and tail of the guide tube **114** in the camera coordinate system. Typically, information about the position of the end effector **112** from such an array **612** and information about the location of a target trajectory from another tracked source are used to calculate the required moves that must be input for each axis of the robot **102** that will move the guide tube **114** into alignment with the trajectory and move the tip to a particular location along the trajectory vector.

[0122] Sometimes, the desired trajectory is in an awkward or unreachable location, but if the guide tube **114** could be swiveled, it could be reached. For example, a very steep trajectory pointing away from the base **106** of the robot **102** might be reachable if the guide tube **114** could be swiveled upward beyond the limit of the pitch (wrist up-down angle) axis, but might not be reachable if the guide tube **114** is attached parallel to the plate connecting it to the end of the wrist. To reach such a trajectory, the base **106** of the robot **102** might be moved or a different end effector **112** with a different guide tube attachment might be exchanged with the working end effector. Both of these solutions may be time consuming and cumbersome.

[0123] As best seen in FIGS. **14A** and **14B**, if the array **908** is configured such that one or more of the markers **918A-918D** are not in a fixed position and instead, one or more of the markers **918A-918D** can be adjusted, swiveled, pivoted, or moved, the robot **102** can provide updated information about the object being tracked without disrupting the detection and tracking process. For example, one of the markers **918A-918D** may be fixed in position and the other markers **918A-918D** may be moveable; two of the markers **918A-918D** may be fixed in position and the other markers **918A-918D** may be moveable; three of the markers **918A-918D** may be fixed in position and the other marker **918A-918D** may be moveable; or all of the markers **918A-918D** may be moveable.

[0124] In the embodiment shown in FIGS. **14A** and **14B**, markers **918A, 918 B** are rigidly connected directly to a base **906** of the end-effector **912**, and markers **918C, 918D** are rigidly connected to the tube **914**. Similar to array **612**, array **908** may be provided to attach the markers **918A-918D** to the end-effector **912**, instrument **608**, or other object to be tracked. In this case, however, the array **908** is comprised of a plurality of separate components. For example, markers **918A, 918B** may be connected to the base **906** with a first array **908A**, and markers **918C, 918D** may be connected to the guide tube **914** with a second array **908B**. Marker **918A** may be affixed to a first end of the first array **908A** and marker **918B** may be separated a linear distance and affixed to a second end of the first array **908A**. While first array **908** is substantially linear, second array **908B** has a bent or V-shaped configuration, with respective root ends, connected to the guide tube **914**, and diverging therefrom to distal ends in a V-shape with marker **918C** at one distal end and marker **918D** at the other distal end. Although specific configurations are exemplified herein, it will be appreciated that other asymmetric designs including different numbers and types of arrays

908A, 908B and different arrangements, numbers, and types of markers **918A-918D** are contemplated.

[0125] The guide tube **914** may be moveable, swivelable, or pivotable relative to the base **906**, for example, across a hinge **920** or other connector to the base **906**. Thus, markers **918C, 918D** are moveable such that when the guide tube **914** pivots, swivels, or moves, markers **918C, 918D** also pivot, swivel, or move. As best seen in FIG. **14A**, guide tube **914** has a longitudinal axis **916** which is aligned in a substantially normal or vertical orientation such that markers **918A-918D** have a first configuration. Turning now to FIG. **14B**, the guide tube **914** is pivoted, swiveled, or moved such that the longitudinal axis **916** is now angled relative to the vertical orientation such that markers **918A-918D** have a second configuration, different from the first configuration.

[0126] In contrast to the embodiment described for FIGS. **14A-14D**, if a swivel existed between the guide tube **914** and the arm **104** (e.g., the wrist attachment) with all four markers **918A-918D** remaining attached rigidly to the guide tube **914** and this swivel was adjusted by the user, the robotic system **100, 300, 600** would not be able to automatically detect that the guide tube **914** orientation had changed. The robotic system **100, 300, 600** would track the positions of the marker array **908** and would calculate incorrect robot axis moves assuming the guide tube **914** was attached to the wrist (the robot arm **104**) in the previous orientation. By keeping one or more markers **918A-918D** (e.g., two markers **918C, 918D**) rigidly on the tube **914** and one or more markers **918A-918D** (e.g., two markers **918A, 918B**) across the swivel, automatic detection of the new position becomes possible and correct robot moves are calculated based on the detection of a new tool or end-effector **112, 912** on the end of the robot arm **104**.

[0127] One or more of the markers **918A-918D** are configured to be moved, pivoted, swiveled, or the like according to any suitable means. For example, the markers **918A-918D** may be moved by a hinge **920**, such as a clamp, spring, lever, slide, toggle, or the like, or any other suitable mechanism for moving the markers **918A-918D** individually or in combination, moving the arrays **908A, 908B** individually or in combination, moving any portion of the end-effector **912** relative to another portion, or moving any portion of the tool **608** relative to another portion.

[0128] As shown in FIGS. **14A** and **14B**, the array **908** and guide tube **914** may become reconfigurable by simply loosening the clamp or hinge **920**, moving part of the array **908A, 908B** relative to the other part **908A, 908B**, and retightening the hinge **920** such that the guide tube **914** is oriented in a different position. For example, two markers **918C, 918D** may be rigidly interconnected with the tube **914** and two markers **918A, 918B** may be rigidly interconnected across the hinge **920** to the base **906** of the end-effector **912** that attaches to the robot arm **104**. The hinge **920** may be in the form of a clamp, such as a wing nut or the like, which can be loosened and retightened to allow the user to quickly switch between the first configuration (FIG. **14A**) and the second configuration (FIG. **14B**).

[0129] The cameras **200, 326** detect the markers **918A-918D**, for example, in one of the templates identified in FIGS. **14C** and **14D**. If the array **908** is in the first configuration (FIG. **14A**) and tracking cameras **200, 326** detect the markers **918A-918D**, then the tracked markers match Array Template **1** as shown in FIG. **14C**. If the array **908** is the second configuration (FIG. **14B**) and tracking cameras **200, 326** detect the same markers **918A-918D**, then the tracked markers match Array Template **2** as shown in FIG. **14D**. Array Template **1** and Array Template **2** are recognized by the system **100, 300, 600** as two distinct tools, each with its own uniquely defined spatial relationship between guide tube **914**, markers **918A-918D**, and robot attachment. The user could therefore adjust the position of the end-effector **912** between the first and second configurations without notifying the system **100, 300, 600** of the change and the system **100, 300, 600** would appropriately adjust the movements of the robot **102** to stay on trajectory.

[0130] In this embodiment, there are two assembly positions in which the marker array matches unique templates that allow the system **100, 300, 600** to recognize the assembly as two different tools or two different end effectors. In any position of the swivel between or outside of these two

positions (namely, Array Template 1 and Array Template 2 shown in FIGS. 14C and 14D, respectively), the markers **918A-918D** would not match any template and the system **100, 300, 600** would not detect any array present despite individual markers **918A-918D** being detected by cameras **200, 326**, with the result being the same as if the markers **918A-918D** were temporarily blocked from view of the cameras **200, 326**. It will be appreciated that other array templates may exist for other configurations, for example, identifying different instruments **608** or other end-effectors **112, 912**, etc.

[0131] In the embodiment described, two discrete assembly positions are shown in FIGS. **14A** and **14B**. It will be appreciated, however, that there could be multiple discrete positions on a swivel joint, linear joint, combination of swivel and linear joints, pegboard, or other assembly where unique marker templates may be created by adjusting the position of one or more markers **918A-918D** of the array relative to the others, with each discrete position matching a particular template and defining a unique tool **608** or end-effector **112, 912** with different known attributes. In addition, although exemplified for end effector **912**, it will be appreciated that moveable and fixed markers **918A-918D** may be used with any suitable instrument **608** or other object to be tracked.

[0132] When using an external 3D tracking system **100, 300, 600** to track a full rigid body array of three or more markers attached to a robot's end effector **112** (for example, as depicted in FIGS. **13A** and **13B**), it is possible to directly track or to calculate the 3D position of every section of the robot **102** in the coordinate system of the cameras **200, 326**. The geometric orientations of joints relative to the tracker are known by design, and the linear or angular positions of joints are known from encoders for each motor of the robot **102**, fully defining the 3D positions of all of the moving parts from the end effector **112** to the base **116**. Similarly, if a tracker were mounted on the base **106** of the robot **102** (not shown), it is likewise possible to track or calculate the 3D position of every section of the robot **102** from base **106** to end effector **112** based on known joint geometry and joint positions from each motor's encoder.

[0133] In some situations, it may be desirable to track the positions of all segments of the robot **102** from fewer than three markers **118** rigidly attached to the end effector **112**. Specifically, if a tool **608** is introduced into the guide tube **114**, it may be desirable to track full rigid body motion of the robot **902** with only one additional marker **118** being tracked.

[0134] Turning now to FIGS. **15A-15E**, an alternative version of an end-effector **1012** having only a single tracking marker **1018** is shown. End-effector **1012** may be similar to the other end-effectors described herein, and may include a guide tube **1014** extending along a longitudinal axis **1016**. A single tracking marker **1018**, similar to the other tracking markers described herein, may be rigidly affixed to the guide tube **1014**. This single marker **1018** can serve the purpose of adding missing degrees of freedom to allow full rigid body tracking and/or can serve the purpose of acting as a surveillance marker to ensure that assumptions about robot and camera positioning are valid.

[0135] The single tracking marker **1018** may be attached to the robotic end effector **1012** as a rigid extension to the end effector **1012** that protrudes in any convenient direction and does not obstruct the surgeon's view. The tracking marker **1018** may be affixed to the guide tube **1014** or any other suitable location of on the end-effector **1012**. When affixed to the guide tube **1014**, the tracking marker **1018** may be positioned at a location between first and second ends of the guide tube **1014**. For example, in FIG. **15A**, the single tracking marker **1018** is shown as a reflective sphere mounted on the end of a narrow shaft **1017** that extends forward from the guide tube **1014** and is positioned longitudinally above a mid-point of the guide tube **1014** and below the entry of the guide tube **1014**. This position allows the marker **1018** to be generally visible by cameras **200, 326** but also would not obstruct vision of the surgeon **120** or collide with other tools or objects in the vicinity of surgery. In addition, the guide tube **1014** with the marker **1018** in this position is designed for the marker array on any tool **608** introduced into the guide tube **1014** to be visible at the same time as the single marker **1018** on the guide tube **1014** is visible.

[0136] As shown in FIG. **15B**, when a snugly fitting tool or instrument **608** is placed within the

guide tube **1014**, the instrument **608** becomes mechanically constrained in 4 of 6 degrees of freedom. That is, the instrument **608** cannot be rotated in any direction except about the longitudinal axis **1016** of the guide tube **1014** and the instrument **608** cannot be translated in any direction except along the longitudinal axis **1016** of the guide tube **1014**. In other words, the instrument **608** can only be translated along and rotated about the centerline of the guide tube **1014**. If two more parameters are known, such as (1) an angle of rotation about the longitudinal axis **1016** of the guide tube **1014**; and (2) a position along the guide tube **1014**, then the position of the end effector **1012** in the camera coordinate system becomes fully defined.

[0137] Referring now to FIG. **15C**, the system **100, 300, 600** should be able to know when a tool **608** is actually positioned inside of the guide tube **1014** and is not instead outside of the guide tube **1014** and just somewhere in view of the cameras **200, 326**. The tool **608** has a longitudinal axis or centerline **616** and an array **612** with a plurality of tracked markers **804**. The rigid body calculations may be used to determine where the centerline **616** of the tool **608** is located in the camera coordinate system based on the tracked position of the array **612** on the tool **608**.

[0138] The fixed normal (perpendicular) distance DF from the single marker **1018** to the centerline or longitudinal axis **1016** of the guide tube **1014** is fixed and is known geometrically, and the position of the single marker **1018** can be tracked. Therefore, when a detected distance $D_{sub.D}$ from tool centerline **616** to single marker **1018** matches the known fixed distance DF from the guide tube centerline **1016** to the single marker **1018**, it can be determined that the tool **608** is either within the guide tube **1014** (centerlines **616, 1016** of tool **608** and guide tube **1014** coincident) or happens to be at some point in the locus of possible positions where this distance $D_{sub.D}$ matches the fixed distance $D_{sub.F}$. For example, in FIG. **15C**, the normal detected distance $D_{sub.D}$ from tool centerline **616** to the single marker **1018** matches the fixed distance $D_{sub.F}$ from guide tube centerline **1016** to the single marker **1018** in both frames of data (tracked marker coordinates) represented by the transparent tool **608** in two positions, and thus, additional considerations may be needed to determine when the tool **608** is located in the guide tube **1014**.

[0139] Turning now to FIG. **15D**, programmed logic can be used to look for frames of tracking data in which the detected distance $D_{sub.D}$ from tool centerline **616** to single marker **1018** remains fixed at the correct length despite the tool **608** moving in space by more than some minimum distance relative to the single sphere **1018** to satisfy the condition that the tool **608** is moving within the guide tube **1014**. For example, a first frame **F1** may be detected with the tool **608** in a first position and a second frame **F2** may be detected with the tool **608** in a second position (namely, moved linearly with respect to the first position). The markers **804** on the tool array **612** may move by more than a given amount (e.g., more than 5 mm total) from the first frame **F1** to the second frame **F2**. Even with this movement, the detected distance $D_{sub.D}$ from the tool centerline vector C' to the single marker **1018** is substantially identical in both the first frame **F1** and the second frame **F2**.

[0140] Logistically, the surgeon **120** or user could place the tool **608** within the guide tube **1014** and slightly rotate it or slide it down into the guide tube **1014** and the system **100, 300, 600** would be able to detect that the tool **608** is within the guide tube **1014** from tracking of the five markers (four markers **804** on tool **608** plus single marker **1018** on guide tube **1014**). Knowing that the tool **608** is within the guide tube **1014**, all 6 degrees of freedom may be calculated that define the position and orientation of the robotic end effector **1012** in space. Without the single marker **1018**, even if it is known with certainty that the tool **608** is within the guide tube **1014**, it is unknown where the guide tube **1014** is located along the tool's centerline vector C' and how the guide tube **1014** is rotated relative to the centerline vector C' .

[0141] With emphasis on FIG. **15E**, the presence of the single marker **1018** being tracked as well as the four markers **804** on the tool **608**, it is possible to construct the centerline vector C' of the guide tube **1014** and tool **608** and the normal vector through the single marker **1018** and through the centerline vector C' . This normal vector has an orientation that is in a known orientation relative to

the forearm of the robot distal to the wrist (in this example, oriented parallel to that segment) and intersects the centerline vector C' at a specific fixed position. For convenience, three mutually orthogonal vectors k' , j' , i' can be constructed, as shown in FIG. 15E, defining rigid body position and orientation of the guide tube **1014**. One of the three mutually orthogonal vectors k' is constructed from the centerline vector C' , the second vector j' is constructed from the normal vector through the single marker **1018**, and the third vector i' is the vector cross product of the first and second vectors k' , j' . The robot's joint positions relative to these vectors k' , j' , i' are known and fixed when all joints are at zero, and therefore rigid body calculations can be used to determine the location of any section of the robot relative to these vectors k' , j' , i' when the robot is at a home position. During robot movement, if the positions of the tool markers **804** (while the tool **608** is in the guide tube **1014**) and the position of the single marker **1018** are detected from the tracking system, and angles/linear positions of each joint are known from encoders, then position and orientation of any section of the robot can be determined.

[0142] In some embodiments, it may be useful to fix the orientation of the tool **608** relative to the guide tube **1014**. For example, the end effector guide tube **1014** may be oriented in a particular position about its axis **1016** to allow machining or implant positioning. Although the orientation of anything attached to the tool **608** inserted into the guide tube **1014** is known from the tracked markers **804** on the tool **608**, the rotational orientation of the guide tube **1014** itself in the camera coordinate system is unknown without the additional tracking marker **1018** (or multiple tracking markers in other embodiments) on the guide tube **1014**. This marker **1018** provides essentially a “clock position” from -180° to $+180^\circ$ based on the orientation of the marker **1018** relative to the centerline vector C' . Thus, the single marker **1018** can provide additional degrees of freedom to allow full rigid body tracking and/or can act as a surveillance marker to ensure that assumptions about the robot and camera positioning are valid.

[0143] FIG. 16 is a block diagram of a method **1100** for navigating and moving the end-effector **1012** (or any other end-effector described herein) of the robot **102** to a desired target trajectory. Another use of the single marker **1018** on the robotic end effector **1012** or guide tube **1014** is as part of the method **1100** enabling the automated safe movement of the robot **102** without a full tracking array attached to the robot **102**. This method **1100** functions when the tracking cameras **200**, **326** do not move relative to the robot **102** (i.e., they are in a fixed position), the tracking system's coordinate system and robot's coordinate system are co-registered, and the robot **102** is calibrated such that the position and orientation of the guide tube **1014** can be accurately determined in the robot's Cartesian coordinate system based only on the encoded positions of each robotic axis.

[0144] For this method **1100**, the coordinate systems of the tracker and the robot must be co-registered, meaning that the coordinate transformation from the tracking system's Cartesian coordinate system to the robot's Cartesian coordinate system is needed. For convenience, this coordinate transformation can be a 4×4 matrix of translations and rotations that is well known in the field of robotics. This transformation will be termed T_{cr} to refer to “transformation—camera to robot”. Once this transformation is known, any new frame of tracking data, which is received as x, y, z coordinates in vector form for each tracked marker, can be multiplied by the 4×4 matrix and the resulting x, y, z coordinates will be in the robot's coordinate system. To obtain T_{cr} , a full tracking array on the robot is tracked while it is rigidly attached to the robot at a location that is known in the robot's coordinate system, then known rigid body methods are used to calculate the transformation of coordinates. It should be evident that any tool **608** inserted into the guide tube **1014** of the robot **102** can provide the same rigid body information as a rigidly attached array when the additional marker **1018** is also read. That is, the tool **608** need only be inserted to any position within the guide tube **1014** and at any rotation within the guide tube **1014**, not to a fixed position and orientation. Thus, it is possible to determine T_{cr} by inserting any tool **608** with a tracking array **612** into the guide tube **1014** and reading the tool's array **612** plus the single marker **1018** of the

guide tube **1014** while at the same time determining from the encoders on each axis the current location of the guide tube **1014** in the robot's coordinate system.

[0145] Logic for navigating and moving the robot **102** to a target trajectory is provided in the method **1100** of FIG. **16**. Before entering the loop **1102**, it is assumed that the transformation T_{cr} was previously stored. Thus, before entering loop **1102**, in step **1104**, after the robot base **106** is secured, greater than or equal to one frame of tracking data of a tool inserted in the guide tube while the robot is static is stored; and in step **1106**, the transformation of robot guide tube position from camera coordinates to robot coordinates T_{cr} is calculated from this static data and previous calibration data. T_{cr} should remain valid as long as the cameras **200**, **326** do not move relative to the robot **102**. If the cameras **200**, **326** move relative to the robot **102**, and T_{cr} needs to be re-obtained, the system **100**, **300**, **600** can be made to prompt the user to insert a tool **608** into the guide tube **1014** and then automatically perform the necessary calculations.

[0146] In the flowchart of method **1100**, each frame of data collected consists of the tracked position of the DRB **1404** on the patient **210**, the tracked position of the single marker **1018** on the end effector **1014**, and a snapshot of the positions of each robotic axis. From the positions of the robot's axes, the location of the single marker **1018** on the end effector **1012** is calculated. This calculated position is compared to the actual position of the marker **1018** as recorded from the tracking system. If the values agree, it can be assured that the robot **102** is in a known location. The transformation T_{cr} is applied to the tracked position of the DRB **1404** so that the target for the robot **102** can be provided in terms of the robot's coordinate system. The robot **102** can then be commanded to move to reach the target.

[0147] After steps **1104**, **1106**, loop **1102** includes step **1108** receiving rigid body information for DRB **1404** from the tracking system; step **1110** transforming target tip and trajectory from image coordinates to tracking system coordinates; and step **1112** transforming target tip and trajectory from camera coordinates to robot coordinates (apply T_{cr}). Loop **1102** further includes step **1114** receiving a single stray marker position for robot from tracking system; and step **1116** transforming the single stray marker from tracking system coordinates to robot coordinates (apply stored T_{cr}). Loop **1102** also includes step **1118** determining current location of the single robot marker **1018** in the robot coordinate system from forward kinematics. The information from steps **1116** and **1118** is used to determine step **1120** whether the stray marker coordinates from transformed tracked position agree with the calculated coordinates being less than a given tolerance. If yes, proceed to step **1122**, calculate and apply robot move to target x, y, z and trajectory. If no, proceed to step **1124**, halt and require full array insertion into guide tube **1014** before proceeding; step **1126** after array is inserted, recalculate T_{cr} ; and then proceed to repeat steps **1108**, **1114**, and **1118**.

[0148] This method **1100** has advantages over a method in which the continuous monitoring of the single marker **1018** to verify the location is omitted. Without the single marker **1018**, it would still be possible to determine the position of the end effector **1012** using T_{cr} and to send the end-effector **1012** to a target location but it would not be possible to verify that the robot **102** was actually in the expected location. For example, if the cameras **200**, **326** had been bumped and T_{cr} was no longer valid, the robot **102** would move to an erroneous location. For this reason, the single marker **1018** provides value with regard to safety.

[0149] For a given fixed position of the robot **102**, it is theoretically possible to move the tracking cameras **200**, **326** to a new location in which the single tracked marker **1018** remains unmoved since it is a single point, not an array. In such a case, the system **100**, **300**, **600** would not detect any error since there would be agreement in the calculated and tracked locations of the single marker **1018**. However, once the robot's axes caused the guide tube **1012** to move to a new location, the calculated and tracked positions would disagree and the safety check would be effective.

[0150] The term "surveillance marker" may be used, for example, in reference to a single marker that is in a fixed location relative to the DRB **1404**. In this instance, if the DRB **1404** is bumped or otherwise dislodged, the relative location of the surveillance marker changes and the surgeon **120**

can be alerted that there may be a problem with navigation. Similarly, in the embodiments described herein, with a single marker **1018** on the robot's guide tube **1014**, the system **100, 300, 600** can continuously check whether the cameras **200, 326** have moved relative to the robot **102**. If registration of the tracking system's coordinate system to the robot's coordinate system is lost, such as by cameras **200, 326** being bumped or malfunctioning or by the robot malfunctioning, the system **100, 300, 600** can alert the user and corrections can be made. Thus, this single marker **1018** can also be thought of as a surveillance marker for the robot **102**.

[0151] It should be clear that with a full array permanently mounted on the robot **102** (e.g., the plurality of tracking markers **702** on end-effector **602** shown in FIGS. 7A-7C) such functionality of a single marker **1018** as a robot surveillance marker is not needed because it is not required that the cameras **200, 326** be in a fixed position relative to the robot **102**, and Tcr is updated at each frame based on the tracked position of the robot **102**. Reasons to use a single marker **1018** instead of a full array are that the full array is more bulky and obtrusive, thereby blocking the surgeon's view and access to the surgical field **208** more than a single marker **1018**, and line of sight to a full array is more easily blocked than line of sight to a single marker **1018**.

[0152] Turning now to FIGS. 17A-17B and 18A-18B, instruments **608**, such as implant holders **608B, 608C**, are depicted which include both fixed and moveable tracking markers **804, 806**. The implant holders **608B, 608C** may have a handle **620** and an outer shaft **622** extending from the handle **620**. The shaft **622** may be positioned substantially perpendicular to the handle **620**, as shown, or in any other suitable orientation. An inner shaft **626** may extend through the outer shaft **622** with a knob **628** at one end. Implant **10, 12** connects to the shaft **622**, at the other end, at tip **624** of the implant holder **608B, 608C** using typical connection mechanisms known to those of skill in the art. The knob **628** may be rotated, for example, to expand or articulate the implant **10, 12**. U.S. Pat. Nos. 8,709,086 and 8,491,659, the disclosures of which are incorporated by reference herein, describe expandable fusion devices and methods of installation.

[0153] When tracking the tool **608**, such as implant holder **608B, 608C**, the tracking array **612** may contain a combination of fixed markers **804** and one or more moveable markers **806** which make up the array **612** or is otherwise attached to the implant holder **608B, 608C**. The navigation array **612** may include at least one or more (e.g., at least two) fixed position markers **804**, which are positioned with a known location relative to the implant holder instrument **608B, 608C**. These fixed markers **804** would not be able to move in any orientation relative to the instrument geometry and would be useful in defining where the instrument **608** is in space. In addition, at least one marker **806** is present which can be attached to the array **612** or the instrument itself which is capable of moving within a pre-determined boundary (e.g., sliding, rotating, etc.) relative to the fixed markers **804**. The system **100, 300, 600** (e.g., the software) correlates the position of the moveable marker **806** to a particular position, orientation, or other attribute of the implant **10** (such as height of an expandable interbody spacer shown in FIGS. 17A-17B or angle of an articulating interbody spacer shown in FIGS. 18A-18B). Thus, the system and/or the user can determine the height or angle of the implant **10, 12** based on the location of the moveable marker **806**.

[0154] In the embodiment shown in FIGS. 17A-17B, four fixed markers **804** are used to define the implant holder **608B** and a fifth moveable marker **806** is able to slide within a pre-determined path to provide feedback on the implant height (e.g., a contracted position or an expanded position). FIG. 17A shows the expandable spacer **10** at its initial height, and FIG. 17B shows the spacer **10** in the expanded state with the moveable marker **806** translated to a different position. In this case, the moveable marker **806** moves closer to the fixed markers **804** when the implant **10** is expanded, although it is contemplated that this movement may be reversed or otherwise different. The amount of linear translation of the marker **806** would correspond to the height of the implant **10**. Although only two positions are shown, it would be possible to have this as a continuous function whereby any given expansion height could be correlated to a specific position of the moveable marker **806**.

[0155] Turning now to FIGS. 18A-18B, four fixed markers **804** are used to define the implant

holder **608C** and a fifth, moveable marker **806** is configured to slide within a pre-determined path to provide feedback on the implant articulation angle. FIG. **18A** shows the articulating spacer **12** at its initial linear state, and FIG. **18B** shows the spacer **12** in an articulated state at some offset angle with the moveable marker **806** translated to a different position. The amount of linear translation of the marker **806** would correspond to the articulation angle of the implant **12**. Although only two positions are shown, it would be possible to have this as a continuous function whereby any given articulation angle could be correlated to a specific position of the moveable marker **806**.

[0156] In these embodiments, the moveable marker **806** slides continuously to provide feedback about an attribute of the implant **10, 12** based on position. It is also contemplated that there may be discreet positions that the moveable marker **806** must be in which would also be able to provide further information about an implant attribute. In this case, each discreet configuration of all markers **804, 806** correlates to a specific geometry of the implant holder **608B, 608C** and the implant **10, 12** in a specific orientation or at a specific height. In addition, any motion of the moveable marker **806** could be used for other variable attributes of any other type of navigated implant.

[0157] Although depicted and described with respect to linear movement of the moveable marker **806**, the moveable marker **806** should not be limited to just sliding as there may be applications where rotation of the marker **806** or other movements could be useful to provide information about the implant **10, 12**. Any relative change in position between the set of fixed markers **804** and the moveable marker **806** could be relevant information for the implant **10, 12** or other device. In addition, although expandable and articulating implants **10, 12** are exemplified, the instrument **608** could work with other medical devices and materials, such as spacers, cages, plates, fasteners, nails, screws, rods, pins, wire structures, sutures, anchor clips, staples, stents, bone grafts, biologics, cements, or the like.

[0158] Turning now to FIG. **19A**, it is envisioned that the robot end-effector **112** is interchangeable with other types of end-effectors **112**. Moreover, it is contemplated that each end-effector **112** may be able to perform one or more functions based on a desired surgical procedure. For example, the end-effector **112** having a guide tube **114** may be used for guiding an instrument **608** as described herein. In addition, end-effector **112** may be replaced with a different or alternative end-effector **112** that controls a surgical device, instrument, or implant, for example.

[0159] The alternative end-effector **112** may include one or more devices or instruments coupled to and controllable by the robot. By way of non-limiting example, the end-effector **112**, as depicted in FIG. **19A**, may comprise a retractor (for example, one or more retractors disclosed in U.S. Pat. Nos. 8,992,425 and 8,968,363) or one or more mechanisms for inserting or installing surgical devices such as expandable intervertebral fusion devices (such as expandable implants exemplified in U.S. Pat. Nos. 8,845,734; 9,510,954; and 9,456,903), stand-alone intervertebral fusion devices (such as implants exemplified in U.S. Pat. Nos. 9,364,343 and 9,480,579), expandable corpectomy devices (such as corpectomy implants exemplified in U.S. Pat. Nos. 9,393,128 and 9,173,747), articulating spacers (such as implants exemplified in U.S. Pat. No. 9,259,327), facet prostheses (such as devices exemplified in U.S. Pat. No. 9,539,031), laminoplasty devices (such as devices exemplified in U.S. Pat. No. 9,486,253), spinous process spacers (such as implants exemplified in U.S. Pat. No. 9,592,082), inflatables, fasteners including polyaxial screws, uniplanar screws, pedicle screws, posted screws, and the like, bone fixation plates, rod constructs and revision devices (such as devices exemplified in U.S. Pat. No. 8,882,803), artificial and natural discs, motion preserving devices and implants, spinal cord stimulators (such as devices exemplified in U.S. Pat. No. 9,440,076), and other surgical devices. The end-effector **112** may include one or instruments directly or indirectly coupled to the robot for providing bone cement, bone grafts, living cells, pharmaceuticals, or other deliverable to a surgical target. The end-effector **112** may also include one or more instruments designed for performing a discectomy, kyphoplasty, vertebrostening, dilation, or other surgical procedure.

[0160] The end-effector itself and/or the implant, device, or instrument may include one or more markers **118** such that the location and position of the markers **118** may be identified in three-dimensions. It is contemplated that the markers **118** may include active or passive markers **118**, as described herein, that may be directly or indirectly visible to the cameras **200**. Thus, one or more markers **118** located on an implant **10**, for example, may provide for tracking of the implant **10** before, during, and after implantation.

[0161] As shown in FIG. **19B**, the end-effector **112** may include an instrument **608** or portion thereof that is coupled to the robot arm **104** (for example, the instrument **608** may be coupled to the robot arm **104** by the coupling mechanism shown in FIGS. **9A-9C**) and is controllable by the robot system **100**. Thus, in the embodiment shown in FIG. **19B**, the robot system **100** is able to insert implant **10** into a patient and expand or contract the expandable implant **10**. Accordingly, the robot system **100** may be configured to assist a surgeon or to operate partially or completely independently thereof. Thus, it is envisioned that the robot system **100** may be capable of controlling each alternative end-effector **112** for its specified function or surgical procedure. Although the robot and associated systems described above are generally described with reference to spine applications, it is also contemplated that the robot system is configured for use in other surgical applications, including but not limited to, surgeries in trauma or other orthopedic applications (such as the placement of intramedullary nails, plates, and the like), cranial, neuro, cardiothoracic, vascular, colorectal, oncological, dental, and other surgical operations and procedures. According to some embodiments discussed below, robot systems may be used for brain surgery applications.

[0162] Current 2-BB-plane methods to register fluoroscopic x-ray images (also referred to as fluoroscopic images, fluoro images, fluoroscopic shots, fluoro shots, fluoroscopic x-ray shots, etc.) to optical tracking cameras when using fluoro-only or pre-op computerized tomography (CT) scan workflows in surgical navigation may have shortcomings as discussed below. [0163] 1) It may be difficult to automatically and/or accurately locate shadows of metal BBs of a fluoroscopic fixture in the presence of other shadows on fluoroscopic x-ray images. [0164] 2) The presence of multiple BBs or other radio-opaque fiducials on fluoroscopic x-ray images may cause undesirable clutter on the images. [0165] 3) The height of the outer BB plane may make it difficult to get the fluoroscopic x-ray detector close enough to the patient, causing fluoroscopic x-ray images to be too magnified (e.g., with a field of view that is too narrow). [0166] 4) Because the registration algorithm may require several BBs to be accurately found on the fluoroscopic x-ray images, it may not be possible to collimate the images to the desired extent.

[0167] According to some embodiments of inventive concepts, some or all of the above issues may be addressed.

[0168] Surgical navigation often uses/requires the registration of a pair of fluoroscopic x-ray images to an optical tracking system used to provide robotic control/navigation and/or surgical planning. For example, the current ExcelsiusGPS application allows planning of screw placements, positioning of the robot, and/or navigation of tools based on two registered fluoroscopic x-ray image shots alone (fluoro workflow). The current ExcelsiusGPS application also allows co-registration of a pair of registered fluoroscopic x-ray image shots to a prior CT scan (pre-operative CT workflow). In both workflows, accurate registration of the pair of fluoro images to the optical tracking cameras may be necessary.

[0169] A C-arm fluoroscopic x-ray imaging system (also referred to as a fluoroscopy unit/system or fluoroscopic unit/system) is illustrated in FIG. **22**. As shown, an x-ray source **3001** (also referred to as a source, an emitter, an x-ray emitter, etc.) and an x-ray detector **3005** (also referred to as a detector, collector, or image intensifier) are mounted on a C-arm **3009**. The C-arm **3009** may be mounted to a base **3011** of the system via a positioning mechanism **3015** that is configured to provide in-out Tx and up-down Ty translational movement and to provide tilt Rx, wig-wag Ry, and orbit Rz rotational movement of the C-arm **3009** relative to the base **3011**. By positioning the

source **3001** and detector **3005** on opposite sides of the patient (e.g., a patient lying on an operating table), x-ray radiation can be transmitted by source **3001** through the patient to detector **3005**, and the detected radiation from detector **3005** can be processed to generate a fluoroscopic x-ray image of the patient's anatomy (also referred to as a fluoro shot), and the C-arm **3009** can be rotated about tilt Rx, wig-wag Ry, and/or orbit Rz so that multiple images of the same anatomy/tissue may be generated from different angles of tilt Rx, wig-wag Ry, and/or orbit Rz. A 3-D reference tracking volume may be determined using two images taken from different angles of tilt Rx, wig-wag Ry, and/or orbit Rz, and this 3-D reference tracking volume may be used to predict how objects in the path of x-rays would appear on the images, and vice versa. That is, by tracking the positions of the collector and emitter when x-ray shots are taken and detecting shadows of reference fiducials in these x-ray shots, there is a closed-form mathematical solution that allows the position of any new object appearing in the 3-D reference volume (defined in camera coordinates) to be mapped to the 2-D shots, and the position of any object simultaneously appearing in both of the 2-D shots to be mapped to the 3-D reference space of the cameras.

[0170] In the fluoro workflow, the registration of the tracked reference space to the x-ray images is all that is used or needed to allow planning and navigation, where screws or other surgical implants are planned on the 2-D shots and tool and robot positions can be driven or observed where they appear on the 2-D shot. It is often useful to register the 3-D reference space described above to the 3-D image volume of a CT scan, cone-beam CT volume, or MRI volume. Once reference space and image volume are registered, planning and navigation of surgical implants can be performed relative to the image volume in 3-D. Some embodiments disclosed herein, relate to registration of a tracked 3-D reference space to the 2-D fluoro shots.

[0171] One method to register a pair of fluoro shots to an optical tracking system is to utilize a fixture **21001** (also referred to as a fluoro fixture) that is attached to the detector **3005** (also referred to as an image intensifier) of the fluoroscopy system of FIG. 22. As shown in FIG. 20, this fixture **21001** has optical tracking markers **21005** (e.g. reflective spheres or light-emitting diodes) used by the optical tracking system to track its position relative to the patient, and the fixture **21001** has multiple arrays of metal spheres (BBs) or other radio-opaque fiducials that are dispersed/embedded in at least two planes to create a pattern of x-ray shadows on the fluoroscopic images/shots. In the example of FIG. 20, a first BB plane **21009a** is spaced apart from the detector **3005**, and a second BB plane **21009b** is mounted adjacent the detector **3005** so that the second BB plane **21009b** is between the first BB plane **21009a** and the detector **3005**. The BBs of FIG. 20 are not visible, but are embedded in a radiolucent material comprising the respective BB planes so that each BB blocks x-rays to provide a respective shadow in a fluoroscopic image generated using detector **3005**. The positions of the fluoroscopic fixture's optical tracking markers **21005** are captured by the optical tracking system at the time the fluoroscopic shot/image is taken, allowing the location of the image plane of detector **3005** to be accurately tracked. From image processing, the shadows created by the BBs and cast on the image plane allow the location of the source **3001** to be accurately determined using the geometric constraints of a pinhole camera model as shown in FIG. 21.

[0172] FIG. 20 illustrates a Globus fluoroscopic registration fixture **21001**. The fixture **21001** clamps on to the x-ray detector **3005** (e.g., image intensifier) of a C-arm fluoroscopy system. Six reflective spheres **21005** are used to provide optical tracking by the optical tracking system and two BB planes **21009a** and **21009b** comprised of black plastic (with **161** BBs embedded therein) are shown.

[0173] FIG. 21 is a schematic diagram illustrating concepts of a pinhole camera model. X-rays from a point source (i.e., x-ray source **3001**, also referred to as an x-ray emitter, an emitter, or a source) travel through tissue (between x-ray source **3001** and 1.sup.st BB plane **21009a**) until reaching the detector **3005** (image intensifier), where the image is generated. If the spacing between BBs and planes is known, the pattern of the x-ray shadows projected on the 2D image plane of detector **3005** dictates where the x-ray source **3001** must have been located relative to the

detector **3005** to create the shot.

[0174] With the location of the source **3001** and detector **3005** accurately determined in camera space (of the optical tracking system), it is possible to exactly define the mathematical transformations to move back and forth between camera space (of the optical tracking system) and the 2D image planes (one plane of each of the pair of fluoroscopic images). For example, if an object is in a known location in camera space, its projection's representation can be rendered onto each image plane exactly where the object's projections would appear as if a new pair of fluoroscopic images/shots were taken from the same orientations of the C-arm **3009**, source **3001**, and detector **3005**. Conversely, if an object's projections are added to the two 2D image planes (like a projected screw image added to the surgical plan on anteroposterior (A-P) and lateral fluoroscopic images/shots), the 3D location of the object in camera space can be exactly determined. Note that there are constraints on where an object's projected image can be placed on the two 2D image planes simultaneously to be in a valid location that would correctly project simultaneously on both images. Consider the analogy of looking at a point in space from front and side perspectives. From a front perspective, it is unclear how far toward the front or back that point is located, but it is certain where from left to right and up to down the point is located. Looking at the same point from the side, it is unclear how far toward the left or right of the 3D (3-dimensional) space that point is, but it is certain where from front to back and up to down the point is located. If moving the 2D (2-dimensional) position of the point as viewed from the front perspective, it is allowed to go anywhere from left to right and the point location would be unchanged from the side perspective, but it cannot be at a different position up to down unless the position of the point on the side perspective changes its up-down location to match.

[0175] Although the methods described above for using the tracked fixture **21001** of FIG. **20** and geometric constraints are effective in accurately determining the locations of the detector **3005** image plane and source **3001** in optical camera space to enable registration of fluoroscopic images/shots to the optical camera space, there may be shortcomings with this method. For example, it may be difficult to automatically locate or accurately locate the shadows of metal BBs (or other fiducials) of the fluoroscopic fixture **21001** in the presence of other shadows on the fluoroscopic image caused, for example, by metallic surgical implants. Some fluoroscopic images/shots may have dark regions of comparable contrast to the metal in the BBs or other fiducials. In such cases, the algorithm may be unable to find or use these fiducial shadows for registration and the accuracy of registration may be reduced or registration may not be possible.

[0176] Another shortcoming of methods using fixture **21001** is that the many BBs or other radio-opaque fiducials on fluoroscopic images must be present for registration, but the BBs may be undesirable to the surgeon who is using the images as a visual aid for surgery. For example, the BBs or other fiducials may obscure part of an anatomical structure that the surgeon needs to see clearly.

[0177] A further shortcoming of methods using fixture **21001** of FIG. **20** is that it may be difficult/impossible to collimate the fluoroscopic images if they are to be used in registration. Collimation is where lead shields are moved in from one or more sides of the x-ray source **3001** to prevent x-rays from reaching outer portions of the collector **3005** and appearing on the fluoroscopic image. Collimation may be useful when some of the x-rays go through tissue but other x-rays go through air only. For example, when taking a lateral fluoroscopic image/shot of a patient, the area to be visualized (torso) is very wide and deep but very short in the anteroposterior (A-P) direction (top to bottom). It may therefore be useful to collimate the x-ray image from the top (or top and bottom) to reduce/prevent the sharp change in image contrast between tissue and air that would otherwise be seen. However, the fluoroscopic images/shots for registration rely on being able to visualize enough fiducial BBs in each shot, and collimation would block some of the BBs.

[0178] A further shortcoming of methods using fixture **21001** of FIG. **20** is that the two planes of BBs that are used/required by the pinhole camera model to localize the source **3001** are bulky and

obtrusive. Although one plane of BBs is flush against the face of x-ray detector **3005**, the second plane typically must protrude outward at least 100 mm from the face of x-ray detector **3005** to achieve the desired accuracy in source **3001** localization. This protrusion of the outer BB plane may make it difficult to get the fluoroscopic unit close enough to the patient. The farther the fluoroscopic unit is from the patient, the more magnified the fluoroscopic image appears due to parallax (resulting in a narrower field of view). Because of this limitation, the fluoroscopic image may be more magnified than the surgeon wants, encompassing, for example, only one spinal level instead of two.

[0179] All of the issues discussed above stem from the usage of two planes of BBs in the calibration algorithm, and so it may be beneficial to eliminate the need for multiple planes of BBs. Instead of using detected BBs in the above method to determine locations of the detector **3005** image plane and source **3001**, the position of the source **3001** may be extrapolated based on the location of an optical tracker mounted to the detector **3005** (image intensifier). However, the source **3001** position cannot be accurately extrapolated because the C-arm **3009** of the fluoroscopy system may flex a small amount when it is in different orientations since the x-ray detector **3005** (image intensifier or flat panel), source **3001** (emitter/anode), and C-arm **3009** itself are somewhat heavy pieces of equipment. Due to flexing of the C-arm **3009**, the location of the source **3001** relative to the detector **3005** may vary by several millimeters between typical positions of the C-arm **3009** such as anteroposterior A-P and lateral, creating an unacceptable inaccuracy in the predicted location of the source **3001** that is used for the registration.

[0180] According to some embodiments of inventive concepts, instead of using projections of fixture fiducials and geometric constraints of a pinhole camera model to determine the location of the x-ray source **3001**, the current monitored orientation of the C-arm **3009** is used to accurately predict how the C-arm **3009** flexes and therefore accurately determine where the source **3001** is located. There are several different ways to monitor the orientation of the C-arm **3009** and several ways to use this information to determine a location of source **3001**, as will be discussed below.

[0181] One way to monitor orientation of C-arm **3009** is to read the values of the C-arm axes automatically or manually and to provide these values to an algorithm that utilizes them to determine the effect on source **3001** location. Typically, C-arm **3009** has three rotational movements: orbit (Rz), tilt (Rx), and wig-wag Ry as shown in FIG. 22. There are additionally some linear translational axes on C-arm **3009**. Since the wig-wag axis Ry is nearly exactly aligned with gravity it can be assumed that there will be no difference in how the C-arm **3009** flexes due to change in wig-wag Ry. Also, the linear translational positions Tx and Ty should not affect how the C-arm **3009** flexes. Therefore, changes in source **3001** position can be characterized relative to detector **3005** position knowing primarily the changes in orbit Rz and tilt Rx.

[0182] FIG. 22 illustrates a configuration of a fluoroscopy system showing different movement axes, including tilt Rx, In-Out Tx, Up-Down Ty, Wig-Wag Ry, and Orbit Rz. The axis about which Wig-Wag occurs (shown as Ry) is typically aligned with gravity unless operating the C-arm **3009** on a slope, and therefore only orbit Rz and tilt Rx should significantly affect how the C-arm **3009** flexes under normal conditions.

[0183] Older C-arm fluoroscopy systems have locking and unlocking mechanisms to allow orbit Rz and tilt Rx to be changed manually, and have a scale indicating within ± 1 degree the position of each axis. A software user interface could therefore allow these values to be entered by the user each time a fluoroscopic shot is taken and would use these values to calculate offset of the source **3001** relative to detector **3005**. Alternately, axes of the C-arm **3009** could be fitted with optical, magnetic, or other encoders to automatically detect the position of each axis and feed this information to the navigation system for processing.

[0184] Newer C-arm fluoroscopy systems have built-in encoders to record the position of each axis digitally. These systems store the axis coordinates to onboard memory and additionally write the coordinates in the DICOM (Digital Imaging and Communication in Medicine) header of each

fluoroscopic image that is taken. Software could therefore use a digital data stream through serial cable or other means between the C-arm fluoroscopy system and navigation system to communicate the current axis positions, especially the orbit Rz and tilt Rx, during fluoroscopic shots. Or, software could retrieve each image/shot as a DICOM file, through Ethernet, serial communication, or wireless data transfer, and read the DICOM tags through standard software methods to retrieve information about the orbit Rz and tilt Rx positions.

[0185] Another way of monitoring the orientation of the C-arm **3009** is to place electronic tilt sensors or an inertial measurement unit (IMU) at some location on the C of the C-arm **3009** to monitor its angular orientation and to feed this information to the navigation system while fluoroscopic shots are taken.

[0186] Two tilt sensors would be used to monitor the offset in the C-arm **3009** corresponding to orbit Rz and tilt Rx. IMUs typically monitor 3 or 6 degrees of freedom and would be oriented when they are mounted on the C-arm **3009** such that the gravitational direction is initially aligned with the vertical axis of the IMU while the C-arm **3009** is in the reference position. Data from tilt sensors and/or IMUs would then be communicated to the navigation system at the time a fluoroscopic shot is taken for registration via Ethernet, WiFi, Bluetooth, serial port, or any suitable means.

[0187] Another way to monitor the orientation of a C-arm **3009** during different shots is to utilize the optical tracking system that is already being used to track the location of the detector **3005** (also referred to as a collector). Doing so would require the system to first define the vertical direction for the system, and to maintain this definition subsequently. For example, a software feature could prompt the user to position the C-arm **3009** with orbit Rz at 0° and tilt Rx at 0° and then hit a button to capture this upright position. Subsequent measurements of the optical tracker on the C-arm **3009** would then be able to determine the change in orientation as long as the cameras of the optical tracking system had not moved. Or, if the optical tracking cameras did move, the vertical direction could still be deduced if an additional tracked reference fixture that indicates the direction of gravity is also viewed, such as tracking markers affixed to a table or other feature that does not change its orientation relative to gravity during the case.

[0188] A different embodiment to monitor the orientation of the C-arm **3009** during different shots is to utilize a gravity vector including optical tracking elements such as reflective spheres. Such a gravity vector could be positioned in close proximity to the fixture used to track the position of the detector **3005** and would only require an additional two markers, tracked as individual strays. If using reflective spheres, the two spheres forming the gravity vector could be connected to a base by a swiveling feature and weight could be added distal to the second sphere to force the two spheres to hang in the direction of gravity as shown in FIG. 23A, FIG. 23B, FIG. 23C, FIG. 24A, and FIG. 24B. This embodiment may be useful relative to using manual or automatic axis entry and/or relying on the tracked fixture orientation to determine C-arm **3009** position because it is less subject to user error. That is, methods using a gravity vector may be automatic with minor additional equipment used/needed and such methods do not require user input or maintenance of communication of other data streams.

[0189] FIGS. 23A, 23B, and 23C illustrate a gravity vector including two additional reflective tracking markers **4001A** and **4001B** adjacent to or within the space occupied by a 4-marker C-arm tracker (including four reflective tracking markers **4005A**, **4005B**, **4005C**, and **4005D**) that is used to track a position of detector **3005**. Three angular positions of the C-arm **3009** are represented and it is demonstrated how the gravity vector changes its orientation relative to the fixture's coordinate system at the different positions. The gravity vector includes two reflective tracking spheres **4001A** and **4001B** connected by a rigid or non-rigid linkage and connected to the base (frame of the fixture's tracker in this example) by a freely swiveling interface such as string or ball joint. The distal tracking sphere **4001B** may be weighted or an additional weight may be attached distal to the tracking sphere **4001B** to provide/ensure that it hangs vertically in alignment with gravity.

[0190] FIGS. 24A and 24B are photographs of a detector fixture **4000** including a gravity vector with tracking markers **4001A** and **4001B** mounted on a frame **4021** with optical/reflective tracking markers **4005A**, **4005B**, **4005C**, and **4005D**. The swiveling feature in this example is a chain **4007** similar to those used for lamp pull cords. This design may be preferred because the chain **4007** is free to rotate about its long axis while also bending at linkage points, and is therefore not likely to snag or get hung up in moving to a new orientation. A fishing weight **4009** was also attached distal to the distal tracking sphere **4001B** of the gravity vector. As discussed below with respect to FIG. 27A, the detector fixture may be permanently mounted to the detector **3005** and/or to a portion of the C-arm **3009** adjacent to detector **3005**. For example, the detector fixture may be mounted to the detector **3005** and/or to a portion of the C-arm **3009** in a manner that the detector fixture is outside a path of x-rays between the source **3001** and detector **3005**.

[0191] One potential issue with a gravity vector such as is described above is that the two spheres connected by a swivel joint may continue to swing for an undesirably long time after moving the C-arm **3009** into a new position, forcing the user to waste time before continuing the registration process. According to some embodiments, damping the movement of the gravity vector may be achieved by connecting the spheres **4001A** and **4001B** to the base using a fluid-filled ball joint with small pores or channels in features of the joint that allow joint movement but only at the rate that fluid (e.g., water or oil) can flow through the pores. According to some other embodiments, a series of two perpendicular damped hinge joints, one corresponding to orbit Rz and one corresponding to tilt Rx, could be used to achieve the same behavior. Each hinge joint could be damped using known methods such as a dashpot across the joint like the mechanisms that control the rate at which household doors close.

[0192] Based on the detected orientation of the C-arm **3009** by any of the techniques described above at the time a fluoroscopic shot was taken, the system would utilize a calibration to determine the location of the source **3001**. Such a calibration could be done at the time the fluoroscopic tracker is mounted on the fluoroscopic unit. In some embodiments, a series of operations would be followed to calibrate and then determine precise locations of source **3001** and detector **3005** as discussed below with respect to the flowchart of FIG. 25.

[0193] FIG. 25 is a flow chart showing the operations to establish a calibration to facilitate adjustment to the extrapolated source **3001** location based on knowledge of C-arm **3009** orientation according to some embodiments of inventive concepts. After completing the calibration, the temporary fixtures can be removed and the offset in source **3001** position of any new fluoroscopic shot can be determined from the stored relationship and the tracked gravity vector. According to some embodiments of FIG. 25, fixtures of FIGS. 27A, 27B, and 27C may be used, and the gravity vector of FIG. 27A may be omitted.

[0194] At block **25001**, a permanent fixture **4000** with a tracking array (including markers **4005A-D**) is attached near x-ray detector **3005** (also referred to as a collector) as shown in FIG. 27A, and temporary fixtures **8010** and **8050** with radio-opaque fiducials and tracking arrays are attached near x-ray source **3001** (also referred to as an emitter) and x-ray detector **3005** as shown in FIGS. 27B and 27C.

[0195] At block **25003**, C-arm **3009** is positioned at the reference position, and a tracked reference fluoroscopic shot is taken while capturing the orientation of C-arm **3009**. Accordingly, positions/orientations of x-ray source **3001**, x-ray detector **3005**, and C-arm **3009** are determined based on tracking arrays of fixtures **4000**, **8010**, and **8050** for the reference position.

[0196] At block **25005**, C-arm **3009** is moved to a new orientation within the normal operating range.

[0197] At block **25007**, tracking data from all tracker arrays (i.e., tracking arrays defined by markers **4005A-D**, markers **8075A-D**, and markers **8053A-D**) and an angular orientation of C-arm **3009** are captured.

[0198] At block **25009**, locations of x-ray source **3001** and x-ray detector **3005** are calculated

relative to permanent fixture **4000** based on data from the temporary tracking arrays of fixtures **8010** and **8050**.

[0199] At block **25011**, operations of blocks **25005**, **25007**, and **25009** are repeated until data is collected for orientations of C-arm **3009** taken over an entire operating range of C-arm **3009**.

[0200] At block **25015**, the x-ray source **3001** offset relative to reference x-ray source **3001** position (from block **25003**) is related to a change in orientation of C-arm **3009** relative to the reference orientation of C-arm **3009** (from block **25003**), and the relationships relative to the tracking array of permanent fixture **4000** are stored for x-ray source **3001** localization with future fluoroscopic shots taken.

[0201] FIG. **26** is a Flow chart for specific usage of an embodiment with gravity vector (e.g., using fixture **4000** of FIG. **27A** with the gravity vector). If the method used to determine C-arm orientation is the gravity vector, the flow chart of FIG. **25** can be expressed instead as the flow chart of FIG. **26**.

[0202] At block **26001**, a permanent fixture **4000** with a tracking array (including markers **4005A-D**) and gravity vector (including markers **4001A-B**) is attached near x-ray detector **3005** (also referred to as a collector) as shown in FIG. **27A**, and temporary fixtures **8010** and **8050** with radio-opaque fiducials and tracking arrays are attached near x-ray source **3001** (also referred to as an emitter) and x-ray detector **3005** as shown in FIGS. **27B** and **27C**.

[0203] At block **26003**, C-arm **3009** is positioned at the reference position, and a tracked reference fluoroscopic shot is taken while capturing orientations of C-arm **3009** and gravity vector. Accordingly, positions/orientations of x-ray source **3001**, x-ray detector **3005**, C-arm **3009**, and gravity vector are determined based on tracking arrays of fixtures **4000**, **8010**, and **8050** for the reference position.

[0204] At block **26005**, C-arm **3009** is moved to a new orientation within the normal operating range.

[0205] At block **26007**, tracking data from all tracker arrays (i.e., tracking arrays defined by markers **4005A-D**, markers **8075A-D**, and markers **8053A-D**) and the gravity vector are captured.

[0206] At block **26009**, locations of x-ray source **3001** and x-ray detector **3005** are calculated relative to permanent fixture **4000** based on data from the temporary tracking arrays of fixtures **8010** and **8050**.

[0207] At block **26011**, operations of blocks **26005**, **26007**, and **26009** are repeated until data is collected for orientations of C-arm **3009** taken over an entire operating range of C-arm **3009**.

[0208] At block **26015**, the x-ray source **3001** offset relative to reference x-ray source **3001** position (from block **26003**) is related to a change in orientation of gravity vector relative to the reference gravity vector orientation (from block **26003**), and the relationships relative to the tracking array of permanent fixture **4000** are stored for x-ray source **3001** localization with future fluoroscopic shots taken.

[0209] Example embodiments of the permanent and temporary fixtures that are attached to the C-arm **3009**, detector **3005**, and/or source **3001** that are mentioned in the first operation of FIG. **25** and/or FIG. **26** are shown in FIGS. **27A**, **27B**, and **27C**.

[0210] FIG. **27A** illustrates a permanent detector fixture **4000** according to the structure shown in FIGS. **24A** and **24B** mounted to a portion of C-arm **3009** adjacent an upper surface of detector **3005**, such that detector **3005** is between the detector fixture **4000** and the source **3001**.

Accordingly, the permanent detector fixture **4000** does not obstruct a path of x-rays from source **3001** to detector **3005**.

[0211] FIG. **27B** illustrates a temporary detector fixture **8010** mounted on a face **3006** of detector **3005** between the detector **3005** and source **3001**.

[0212] FIG. **27C** illustrates a temporary source fixture **8050** mounted on source **3001** between source **3001** and detector **3005**. As shown, source fixture **8050** includes a frame **8051** and a plurality of optical tracking markers **8053A**, **8053B**, **8053C**, and **8053D**.

[0213] FIGS. 27A, 27B, and 27C thus illustrate fixtures used to calibrate the fluoroscopic imaging system, based on optical tracking using the optical tracking markers **4005A**, **4005B**, **4005C**, **4005D**, **8053A**, **8053B**, **8053C**, **8053D**, **8075A**, **8075B**, **8075C**, and **8075D** (e.g., provided as optically reflective spheres).

[0214] As shown in FIG. 27A, permanent detector fixture **4000** is attached on/near the detector **3005**. In this example, the permanent detector fixture **4000** also has a gravity vector (including optical tracking markers **4001A** and **4001B** and weight **4009**) attached to its frame **4021**, and optical tracking markers **4005A**, **4005B**, **4005C**, and **4005D** (e.g., provided as reflective spheres).

[0215] As shown in FIG. 27B, temporary fixture **8010** is attached on/near the imaging plane of detector **3005**. Fixture **8010** has a tracking array including a portion **8073A** of frame **8073** that extends perpendicular from face **3006** of detector **3005** and optical tracking markers **8075A**, **8075B**, **8075C**, and **8075D** on frame portion **8073A**. Fixture **8010** also includes a plane of BBs in a portion **8073B** of frame **8073** that is flat against and parallel to the face **3006** of detector **3005** so that the plane of BBs and portion **8073B** of the frame are between detector **3005** and source **3001**. The optically tracking array (including optical tracking markers **8075A**, **8075B**, **8075C**, and **8075D**) is thus in a known/fixed location relative to the BB plane because this tracking array and the BBs of this plane are both affixed to and/or provided in/on the same rigid frame **8073**, thereby allowing the position of the BB plane (and the positions of the BBs therein) to be accurately tracked using the optical tracking array.

[0216] As shown in FIG. 27C, temporary fixture **8050** is attached near the source **3001** (e.g., between the source **3001** and detector **3005**). The fixture **8050** has a tracking array (including optical tracking markers **8053A**, **8053B**, **8053C**, and **8053D**) and also a plane of BBs **8055** (also referred to as an x-ray opaque fiducial pattern) embedded in frame **8051** between source **3001** and detector **3005**. The tracking array (including optical tracking markers **8053A**, **8053B**, **8053C**, and **8053D**) is in a known location relative to the BB plane **8055** because both are affixed to and/or provided in/on the same rigid frame **8051**, thereby allowing the position of the BB plane **8055** (and the positions of the BBs therein) to be accurately tracked using the optical tracking array. The BB plane **8055** is perpendicular to the path of x-rays (from source **3001** to detector **3005**) and a few inches away from the x-ray source **3001**.

[0217] FIG. 28 illustrates an embodiment of the reference fluoroscopic image/shot described in FIG. 25 (operation **25003**) or FIG. 26 (operation **26003**) that could be the standard position for an anteroposterior shot, with detector **3005** above the procedure table and the source **3001** below the procedure table. This position is typically labeled as 0° tilt Rx and 0° orbit Rz in standard fluoroscopic C-arm systems. A pinhole camera model constraint would be applied to the fluoroscopic image/shot taken in the reference position (operation **25003** or **26003**) to accurately locate the source **3001** based on the projections of the BBs from BB plane **8055** of fixture **8050** on source **3001** and based on the projections of BBs from the BB plane of fixture **8010** on detector **3005**. The distance between BB planes, required by the pinhole camera model, would be tracked using optical tracking markers (**8075A**, **8075B**, **8075C**, **8075D**, **8053A**, **8053B**, **8053C**, and **8053D**) on the fixtures **8010** and **8050**. Methods for applying projection constraints to data sets are known in the field and are available through open-source software such as OpenCV (open source computer vision, <http://opencv.org>). Since the reference fluoroscopic shot is for calibration purposes only, the fluoroscopic shot may be taken without anything in the field between fixtures **8010** and **8050**. By taking such an “air shot”, it can be provided/ensured that good image contrast will be achieved and that image processing will be able to detect the BB shadows effectively as shown in FIG. 28.

[0218] FIG. 28 illustrates an example of a fluoroscopic x-ray air image/shot **28000** taken with temporary fixtures **8010** and **8050** respectively attached to detector **3005** and source **3001**. The inside ring of BB shadows **28011** and inner BB shadows **28015** are from the BBs of source fixture **8050** and the outside ring of BB shadows **28017** is from the detector fixture **8010**. The BBs near the source **3001** (included in fixture **8050**) have a diameter that is 32% of the diameter of BBs near

the detector **3005** (included in fixture **8010**) but the BB shadows appear roughly the same size due to parallax during projection. Lighter shadows are from the frame **8073** of fixture **8010**.

[0219] An orientation of the gravity vector (also referred to as a gravity vector value) of FIG. **27A** (including optical tracking markers **4001A** and **4001B**) could be displayed as a unit vector with X,Y, and Z components and the gravity vector value can be updated continually by software to monitor the current orientation of the C-arm **3009**. This real-time gravity vector value could be used as a motion-sensing control feature. If the gravity vector is not static, then the C-arm is still in motion or the gravity vector is still swinging after the last movement. Sensed motion of the gravity vector can therefore be used to reduce/prevent/limit unusable fluoroscopic shots from being taken or acquired by the frame grabber through software. The orientation of the gravity vector (i.e., the gravity vector value) can also be recorded for reference, and this reference value utilized in the same or different surgical cases to help guide the user to a particular angular orientation or angular increment from a previous orientation of the C-arm. For example, if the gravity vector pointed a particular direction for one fluoroscopic shot that was later found to be ineffective in registration and the user wished to try a new C-arm orientation that was exactly 100 different from the previous orientation, the stored value of the gravity vector could help precisely find the new position by showing and comparing the current and previous gravity vector. Software could also display the calculated angular difference between the previous and current gravity vector values, taken as the arccosine of the vector dot product of the two unit vectors.

[0220] In the loop of FIG. **25** or the loop of FIG. **26**, the C-arm **3009** would be swung through its standard operating range of positions while tracking the marker arrays on fixtures of FIGS. **27A**, **27B**, and **27C** and the gravity vector (or C-arm orientation by whatever means is used). It is important that the gravity vector, if used, does not move into any position where it hangs up on itself or another part of the fixture, or is for any reason unable to hang freely with gravity during the range studied. The attachment to the gravity vector may therefore be designed in such a way that it hangs down when in the most common orientation and is 180° inverted in the least common orientation. The least common orientation would likely be with the x-ray detector **3005** positioned under the procedure table. This orientation is unlikely because the tracking array (i.e., including the optical tracking markers **4005A**, **4005B**, **4005C**, and **4005D** in the fixture of FIG. **27A**) on the detector **3005** may not be visible in such an orientation.

[0221] In collecting data for the useable range of orientations of the C-arm **3009**, enough samples should be collected that the relationship between the orientation of C-arm **3009** and source **3001** offset (relative to detector **3005**) is well characterized. The primary data to be sampled are the source **3001** and detector **3005** locations, which were established relative to their tracking arrays in the reference shot. Additional fluoroscopic x-ray shots are not needed because the tracking arrays should equally or more accurately locate the source **3001** and detector **3005** than projection data.

[0222] With sampled data at different orientations of C-arm **3009** (also referred to as C-arm orientation), the relationship between source **3001** offset and C-arm orientation must be established. There are different ways to establish this relationship. One way is to use an interpolative lookup table, where different angular orientations corresponding to orbit Rz and tilt Rx are tabulated along with the source **3001** offset in X, Y, and Z dimensions. Then, when a new fluoroscopic x-ray shot is taken at a particular orientation of C-arm **3009**, the X, Y, and Z offsets are read from the lookup table. If there is no entry for the combination of orbit Rz and tilt Rx used, then the offsets are linearly or nonlinearly interpolated from the nearest available entries.

[0223] A different way to set the relationship between source **3001** offset and C-arm orientation is to make a physics model of the C-arm **3009**. The physics model could be calibrated with material and geometric properties that precisely predict how the C-arm **3009** flexes at different orientations. The data collected when sampling at different orientations would be used to tune or optimize the model such that the model is able to generate accurate outputs when given a new orientation of C-arm **3009** for a new x-ray fluoroscopic shot that was not part of the tuning data set.

[0224] A third way to set the relationship between source **3001** offset and C-arm orientation is to utilize machine learning such as a neural network. The training data set would be the data collected during the sweeps of the C-arm **3009** and the corresponding source **3001** offsets. The accuracy of the neural network model in predicting source **3009** offset based on C-arm orientation would depend on how much data was provided to the neural network in its training.

[0225] Another method to establish the relationship between source **3001** offset and C-arm orientation is to fit a general function of two input parameters to an output that could be X, Y or Z Cartesian coordinate shift in the location of the source. The two input parameters i1 and i2 could be orbit Rz and tilt Rx angles or two components of a gravity vector (since a unit orientation vector has two degrees of freedom). Functions could therefore be written as: [0226] $X=f1(i1, i2)$ [0227] $Y=f2(i1, i2)$ [0228] $Z=f3(i1, i2)$

[0229] Once calibrated, three such equations would fully define source **3001** shift as a function of the C-arm orientation. Sampled data could be used to fit each function through an appropriate computational method such as Levenberg-Marquardt Optimization. The form of the function to fit for such a process could be a sine-cosine relationship, polynomial model, Gaussian mixture model, or general curve with any number of degrees of freedom. In one example of an embodiment, a function could be fit for X, Y, and Z Cartesian coordinates based on i1 and i2 such that

[00001] $X = p1 * i1^2 * i2 + p2 * i2^2 * i1 + p3 * i1^2 + p4 * i2^2 + p5 * i1 * i2 + p6 * i1 + p7 * i2 + p8$
where p1 through p8 are parameters solved through an optimization method, and with each parameter representing a weight of the contribution of a component to the function. Similar functions for Y and Z could be written, each with their own set of weights that are solved separately.

[0230] After completing the calibration procedure and logging the source **3001** offset through a range of C-arm orientations, the temporary fixtures of FIGS. 27B and 27C would be removed, leaving just the permanent fixture of FIG. 27A for actual usage during procedures/cases. During data collection, it could be provided/ensured that the relationship is established in reference to the permanent fixture of FIG. 27A. The offset from the permanent fixture to the temporary fixture on the detector **3005** should be fixed, so the relationship of detector **3005** location to the permanent array of FIG. 27A is easily determined/computed. The offset from the permanent fixture of FIG. 27A to the source **3001** in the upright location would have been captured in the second step of the flow chart of FIG. 25 or FIG. 26 and then the source **3001** offset would have been related to this position in the coordinate system of the permanent fixture of FIG. 27A during data collection to keep the relationship always in terms of the permanent fixture. Alternately, the permanent fixture of FIG. 27A could be added after the calibration is completed and an operation similar to our “registration transfer” described elsewhere could be used to transform the relationship that was previously in terms of the temporary fixture of FIG. 27B to be in terms of the permanent fixture of FIG. 27A, after which temporary fixtures are removed.

[0231] Instead of trying to relate the gravity vector to a particular tracking fixture, or designing tracking fixture attachments to ensure that fixtures always attach relative to each other in a reproducible way, the gravity vector could be stored and accessed as “normalized” values. For normalization, two specific orientations could correspond to reference positions. In one embodiment of the concept of normalization, the upright position of the C-arm **3009** could correspond to (0,0,1), or exact alignment with the Z axis, and the lateral position could correspond to (1,0,0), or closest alignment with the X axis. To achieve normalization of the gravity vector in practice, the C-arm **3009** would be put into the upright position and the gravity vector captured relative to any reference tracking fixture (FIGS. 23A, 23B, 23C, 24A, 24B, or 27A). Then, the C-arm **3009** would be put into the lateral position and the gravity vector value captured relative to that position. In each position, the 3×3 rotation matrix to rotate from upright to that actual orientation would be recorded; that is, R.sub.0 could represent the 3×3 rotation of the gravity vector relative to the fixture in upright position and R.sub.90 could represent the 3×3 rotation of the gravity vector

relative to the fixture in lateral position. If a unit vector g initially aligned with the z-axis $g=(0,0,1)$ is rotated by $R_{\text{sub}.0}$, a new vector $g_{\text{sub}.Z}$ exactly aligned with gravitational Z would be described as:

$$[00002] g_Z = R_0 * g .$$

[0232] Similarly, if the same vector is rotated by R_{90} , a new vector $g_{\text{sub}.X.\text{sub}.a}$ approximately aligned with gravitational X would be described as:

$$[00003] g_{Xa} = R_{90} * g .$$

[0233] The cross product of these vectors (after normalization to a unit vector) would be a unit vector $g_{\text{sub}.Y}$ exactly aligned with gravitational Y:

$$g_{\text{sub}.Y} = g_{\text{sub}.Z} \times g_{\text{sub}.X.\text{sub}.a} .$$

[0234] Crossing $g_{\text{sub}.Y}$ and $g_{\text{sub}.Z}$ gives a unit vector exactly aligned with gravitational X:

$$g_{\text{sub}.X} = g_{\text{sub}.Y} \times g_{\text{sub}.Z} .$$

[0235] Therefore, using these vectors as columns of a 3×3 rotation matrix R gives the rotational transformation from the identity matrix to the normalized gravity vector orientation:

$$R = [g_{\text{sub}.X} \ g_{\text{sub}.Y} \ g_{\text{sub}.Z}] .$$

[0236] The inverse of this matrix $R_{\text{sub}.inv}$, when applied to any gravity vector g_r obtained relative to the tracking fixture, gives a normalized gravity vector:

$$[00004] g_N = R_{\text{inv}} * g_F .$$

[0237] Newer C-arm fluoroscopic systems utilize flat panel x-ray detectors, which create undistorted x-ray images. However, older C-arm fluoroscopic systems, which are still in service in many hospitals, utilize image intensifiers as x-ray detectors. Images created using older C-arm fluoroscopic systems may be subject to a greater amount of s-distortion and pincushion distortion than images created using flat panel x-ray detectors. The level of noise is substantial enough and unpredictable enough that raw images may lack the accuracy needed to detect source **3001** location using the pinhole camera constraints mentioned earlier. Raw images from image-intensifier-based detectors may therefore require an additional step to dewarp the images before processing them to be used in registration for any type of workflow such as fluoro-only or pre-op CT workflow. In a registration method where two planes of BBs are present on the fluoroscopic tracking fixture as discussed above with respect to FIG. **20**, BBs from the plane **21009b** nearest the detector **3005** are used for dewarping by applying a dewarping algorithm. One such dewarping algorithm is a morphing algorithm where the pixels within the triangle formed by three detected BB shadows are stretched/shrunk in two dimensions so that the triangle matches the known shape. In the proposed new registration method, it is also possible to continue to use BBs on the detector **3005** for dewarping. Unlike in the old fixture, only one plane of BBs (instead of two planes) would be needed. This single plane would serve to support dewarping and also provide orientation landmarks to provide/ensure each new image matches the arbitrary screen orientation and offset that may differ from image to image. Also unlike in the old methods, it should be possible to collimate as much as desired/needed since dewarping only occurs within regions bounded by visible BBs. At least 3 orientation BBs would still need to be visible within the collimated region to provide/ensure correct image orientation. Since a BB plane is also used/required for the tracking fixture of FIG. **27B** on the detector **3005** in the calibration process, the temporary tracker of FIG. **27B** on image-intensifier-based detectors **3005** could double as the permanent tracker.

[0238] The fluoroscopic fixture of FIG. **20** using two planes of BBs **21009a** and **21009b** has some features used to account for user-controlled on-screen image rotation, image magnification, and image flip. It is important to account for these factors, which may vary from shot to shot depending on how the user adjusts settings and noise inherent in the system, to get an accurate mapping of the

3D coordinate system to the 2D images and vice versa. To account for magnification, some BBs could be arranged in a ring with known diameter, and spacings between dewarping BBs could be known. To account for on-screen image rotation, the BBs in the existing image-intensifier fixture are laid out in an asymmetrical pattern. The asymmetrical pattern is asymmetrical rotationally and also asymmetrical from front to back, so that a flipped and/or rotated image is immediately recognized as such.

[0239] A new fluoroscopic fixture with a single plane need not utilize any BBs for dewarping when used with flat panel units, and just needs to account for image orientation, flip, and magnification. The existing pattern on the old fixture's far plane **21009a** uses 6 orientation BBs distributed in different rotational positions at a constant radius from fixture center (not shown). Such an arrangement would fulfill the requirements of accounting for orientation, flip, and magnification, but may be somewhat limited for collimated shots. Another possible BB layout could utilize BBs clustered in groups of 3 to allow for collimation from top-bottom or left-right as shown in FIGS. **29A**, **29B**, and **29C**. As shown in FIG. **29A**, fixture **29001** may be provided on the face **3006** of detector **3005** between detector **3005** and source **3001**. Fixture **29001** may include 4 groups of BBs (also referred to as fiducials) **29005A**, **29005B**, **29005C**, and **29005D**, with each group including three BBs, and with each group of 3 BBs provided in a different pattern. In FIG. **29A**, fixture **29001** includes an optical tracking array with optical tracking markers **29011A**, **29011B**, **29011C**, and **29011D** provided in a relationship that is fixed relative to BBs of groups **29005A**, **29005B**, **29005C**, and **29005D**. Fixture **29001** may thus be used like Fixture **8010** of FIG. **27B** for calibration, but because Fixture **29001** is intended to be maintained on detector **3005** during procedures, the tracking array (including tracking markers **29011A**, **29011B**, **29011C**, and **29011D**) is extended to a side of detector **3005** and away from source **3001** as shown in FIG. **29A**.

[0240] Examples of images taken using fixture **29001** of FIG. **29A** with collimation are provided in FIGS. **29B** and **29C**. In FIG. **29B**, shadows **29005B'** and **29005D'** of BB groups **29005B** and **29005D** are present in the image, and the patterns of these BB groups can be used to determine the orientation of this image. In FIG. **29C**, shadows **29005A'** and **29005C'** of BB groups **29005A** and **29005C** are present in the image, and the patterns of these BB groups can be used to determine the orientation of this image.

[0241] Another possible minimal fiducial layout could be a crosshair across the center of the screen with additional crosshairs or spaces and an asymmetrical feature as shown in FIG. **30**. Other possible features with useful/necessary properties of asymmetry and spacing could be provided to detect these factors. In addition to being useful for detecting image rotation, magnification, and/or flip, these fiducials could be re-projected by software onto the image and should overlay perfectly if calibration is correct, giving the user confidence in valid registration. Such a software feature is similar to how the BBs from the two planes in the old fluoroscopic fixture are re-projected onto the images for a check of good registration.

[0242] FIG. **29A** illustrates a configuration of BBs on a fixture **29001** with clusters of 3 BBs in top (**29005D**), bottom (**29005B**), left (**29005A**), and right (**29005C**) regions. Positioning BBs in this way allows 6 BBs to still be easily detected even in highly collimated shots with FIG. **29B** being an x-ray fluoroscopic shot image with left-right collimation and with FIG. **29C** being an x-ray fluoroscopic shot image with top-bottom collimation.

[0243] FIG. **30** illustrates an example of a fiducial configuration of a feature added to the detector fixture **29001** of FIG. **29A** that could appear on fluoroscopic x-ray shots to serve as a reference for the image rotation, translation, magnification, and/or flip. The ring **30001** has a known diameter, and the length of each line segment **30005A**, **30005B**, and **30005C** before the respective gap **30007A**, **30007b**, and **30007C** is also known. The gaps are asymmetrically placed so that it is known rotationally and with regard to flip where the gaps should be on the image. The crosshairs can also be projected onto the image by software, providing feedback to the user on where the calibration believes is the center of the image plane.

[0244] Because each new image captured after calibrating the system is related back to a reference image by transforming (e.g., rotating, scaling, translating, and/or flipping), it is important that the fixture should be attached to the x-ray detector **3005** (image plane) securely. If the fixture were to slip, the source location relative to the image would no longer be accurate, even if adjusted for gravity. One approach would be to collect a new, calibrated reference air shot at the beginning of the procedure/case, along with gravity vector information in upright and lateral positions. The reference air shot would not have any anatomy present and would have a second BB plane present to allow accurate calculation of the source **3001** location using the pinhole camera model. The second plane could be a tracked BB plane such as plane **8055** of the source fixture of FIG. **27C** or a rigidly attached temporary plane that is offset from the primary plane by a known amount. An alternate approach would be to design the detector fixture in such a way that it is known that it will always stay well attached or if it is detached, it can be reattached to exactly the same position.

[0245] Because accuracy relies on the detector fixture **29001** being secure and/or accurately re-attachable to the detector **3005** in a reproducible exact way, it may be desirable to provide a way to know if the calibration is still valid after some time. According to some embodiments, a method to assess validity of the calibration is to attach one or more surveillance markers **8091** and **8093**. In some embodiments, one surveillance marker **8091** is attached on/near the detector **3005** and one surveillance marker **8093** is attached on/near the source **3001** as shown in FIG. **31**. These surveillance markers, for example, may be reflective spheres, reflective stickers, or active markers detectable by the tracking system in the same way as the permanent fixture is detectable/tracked. The locations of these markers would be optically tracked relative to the permanent fixture at the time the calibration is performed. Later, when checking calibration, the surveillance marker **8091** near the detector **3005** should be in the same fixed position relative to the permanent tracking fixture **4000** and/or **8010** as it was during calibration. If the permanent tracking fixture or its tracking array become dislodged or bent, the surveillance marker can be used to detect offset from its predicted position and it will be apparent that there is a problem. Additionally, since the surveillance marker **8093** near the source **3001** should move in the same way as the source **3001** moves with changing C-arm and/or detector **3005** orientation, the system should be able to calculate and predict where the surveillance marker near the source **3001** would move at any C-arm and/or detector **3005** orientation. When checking calibration, the C-arm **3009** could be placed into a few different orientations without shooting x-rays. If the tracked surveillance marker location is not in the predicted location (based on recalled surveillance marker upright position adjusted to calibrated offset) at all tested orientations of the C-arm **3009** and/or detector **3005**, then the calibration is incorrect and needs to be re-run.

[0246] FIG. **31** is a photograph of a C-arm **3009** with fixtures **8050**, **8010**, and **4000** attached and surveillance markers **8091** and **8093** drawn in two locations, one on/near the source **3001** and one on/near the detector **3005**. The surveillance markers would allow the calibration accuracy to be checked before starting a procedure/case.

[0247] The overall workflow when using this fixture after calibration is summarized in the flow chart of FIG. **32**. Every new image that comes in is aligned with a reference image, and has a known shift in source **3001** relative to the reference image, so if the image-to-fixture transformation of the reference image is known or has been previously found using two planes, then a single plane fixture can be used according to this workflow.

[0248] FIG. **32** is a flowchart of an overall workflow when utilizing a calibrated fixture on the day of a procedure/case (e.g., a surgery) according to some embodiments of inventive concepts.

[0249] At block **32001**, fixture **4000** is attached to C-arm **3009** as illustrated in FIG. **31**.

[0250] At block **32005**, an additional operation may be performed if an attachment/alignment of fixture **4000** on C-arm **3009** is not reproducible. If attachment/alignment of fixture **4000** on C-arm **3009** is not reproducible at block **32005**, an anteroposterior A-P fluoroscopic image/shot (with detector **3005** aligned vertically over source **3001**) is taken and a lateral fluoroscopic image/shot

(with detector **3005** and source **3001** aligned horizontally) is taken at block **32009**, and these fluoroscopic shots are used to normalize the gravity vector of fixture **4000**. If attachment/alignment of fixture **4000** is reproducible at block **32009**, the fluoroscopic shots of block **32009** may be omitted.

[0251] At block **32011**, a fluoroscopic image is captured while tracking fixture **4000** and the gravity vector.

[0252] At block **32015**, the image is transformed to match a reference image, accounting for rotation, translation, magnification, and/or flip.

[0253] At block **32019**, a source **3001** offset is found for the detected gravity vector.

[0254] At block **32021**, an image-to-fixture transformation is calculated by adjusting the existing transformation of the reference image based on the source **3001** offset/shift.

[0255] According to some embodiments, a feature may be provided to detect and indicate that x-rays are present. Such a feature is useful because the tracking information related to a shot must be synchronized with the image data in order for a registration to be accurate. Currently, the system detects when the image coming through the video capture board changes, at which time, the tracking data is polled. However, it is possible for false negatives or false positives to occur with new image detection, or for movement of the C-arm **3009** during shot capture to cause inaccuracy.

[0256] According to some embodiments, one mechanism to synchronize imaging and tracking data is to transmit through wireless (e.g., WiFi, Bluetooth, etc.) or wired methods the state of an electronic x-ray detector to the robot system (e.g., to the robot controller **35000** of FIG. **35**). This synchronizing signal could be read through a DIO (digital input-output) port and used by software to ensure that tracking data are only collected when x-rays are active.

[0257] According to some other embodiments, another mechanism to synchronize imaging and tracking data is to use information from an electronic x-ray presence sensor **33019** as a signal that causes one or more tracking markers (e.g., tracking markers of active marker array **33001**) to strobe only when x-rays are present/detected. If all the markers of the array strobe only when x-rays are detected, the system would only be able to track the fixture during x-ray fluoroscopic shots. Or, if only one marker on the array or adjacent to the array is switched on responsive to detecting x-rays, the system could still determine if the fixture is within view of the cameras even though an x-ray shot is not being collected. Such an active marker array **33001** is illustrated in FIG. **33**.

[0258] According to some other embodiments, a passive reflective marker on a pivot or piston (also referred to as a movable marker) could be actuated to indicated when an x-ray fluoroscopic shot is being taken. A motor or other driver would be switched on and would force the pivot or piston to one position when x-rays are present/detected. When x-rays are not present/detected, the motor would switch off and the pivot or piston would return the passive marker under a spring-loaded mechanism to its resting position. The tracking cameras would detect the marker's position and only record the fixture location when x-rays are active. Such a movable marker array could be used instead of the active marker array of **33001** of FIG. **33**.

[0259] According to some embodiments, an electronic version of the detector fixture is illustrated in FIG. **33**. This detector fixture may use an array of active markers **33001** that emit infrared light and can be tracked more accurately than passive reflective spheres. Additionally, the detector fixture may use an IMU **33005** instead of an optically tracked gravity vector to accurately detect the orientation of the C-arm **3009** and/or detector **3005** relative to gravity. The fixture may include on-board memory **33009A**, possibly loadable through a USB flash drive or other portable memory storage device, or loadable through wireless communication, to hold the parameters of a lookup table or function that relates IMU output to an offset (also referred to as a shift) of source **3001** relative to detector **3005**. An identification ID of the fluoroscopic C-arm system being used could be received/retrieved automatically using a near field communication NFC tag, a Quick Response QR code and reader, a bar code, a Radio Frequency Identifier RFID tag, or other type of proximity device mounted on the C-arm detector **3005** near where the fixture attaches to provide/ensure that

the correct parameters are loaded for the C-arm fluoroscopic system to which the detector fixture is attached and to allow the same detector fixture to be used on different fluoroscopic C-arm systems. The detector fixture could be wireless and battery powered, or wired for A/C power. In either case, the active markers of array **33001** could be electronically linked to the rest of the fixture and/or powered using inductive coupling or other methods, thereby allowing the markers to be provided sterile, and attached to array mount **33011** of the fixture over/through the outside of a sterile drape **33015** around the fixture and detector similar to how the end effector of the robot currently attaches to the robot arm over the drape. This fixture may additionally have an x-ray presence sensor **33019** to synchronize tracking data with fluoroscopic x-ray shots.

[0260] The detector fixture of FIG. **33** may also include a BB plane as discussed above for example with respect to the fixture of FIG. **29A**. In general, the fixture of FIG. **33** may be similar to the fixture of FIG. **29A** with the addition/substitution of active components such as IMU **33005**, x-ray presence sensor **33019**, and active marker array **33001**. Moreover, the fixture of FIG. **33** may include the array mount **33011** allowing the detachable coupling of active marker array **33001** through the sterile drape **33015**. [0261] 1) The fixture of FIG. **33** may allow collimation without affecting accuracy of registration as discussed above with respect to FIGS. **29A**, **29B**, and **29C**. As shown in FIG. **33**, the fixture may include fiducials **29000** (also referred to as an x-ray opaque fiducial pattern, e.g., provided using metal BBs or another x-ray opaque material) as discussed above with respect to FIGS. **29A**, **29B**, and **29C**. For example, each grouping (also referred to as a cluster) of 3 fiducials (e.g., provided using BBs) may be provided in a different pattern/orientation to allow identification of each grouping/cluster in a fluoroscopic x-ray image. Moreover, the x-ray opaque fiducial pattern may be provided in/on a layer of an x-ray transparent material, also referred to as a fiducial plane. [0262] 2) The fixture of FIG. **33** may be less obtrusive, allowing it to be positioned closer to the patient, because multiple planes of BBs are not required. [0263] 3) There may be fewer artifacts in the x-ray images produced with the fixture of FIG. **33** by reducing a number of BB shadows (also referred to as fiducial shadows). [0264] 4) Since image processing may be reduced according to some embodiments, x-rays where fiducials cannot be distinguished may pose less of a problem.

[0265] FIGS. **38A** and **38B** are plan views illustrating a fixture **38000** of FIG. **33** according to some embodiments of inventive concepts. In FIGS. **38A** and **38B**, an x-ray opaque fiducial pattern is provided in/on fiducial plane **38011** (also referred to as a BB plane) provided using a layer of an x-ray transparent material, such that all elements of the x-ray opaque fiducial pattern are in a single plane. As shown, the x-ray opaque fiducial pattern may be provided using clusters of BBs (i.e., spherical metal balls, also referred to as ball bearings). According to some other embodiments, the x-ray opaque fiducial pattern may be provided using a cross-hair pattern made up of fiducial segments as discussed above with respect to FIG. **30**. According to still other embodiments, elements of the x-ray opaque fiducial pattern may be provided using circular shaped x-ray opaque discs (e.g., metal discs) with each circular shaped disc having flat upper and lower surfaces, or using a printed/painted pattern of circles (using an x-ray opaque material/paint). For example, the x-ray opaque fiducial pattern may be painted/printed on an upper/lower surface of the fiducial plane **38011**.

[0266] In FIGS. **38A** and **38B**, the array mount of FIG. **33** may be provided using array mounting frame **33011a**, and the active marker array **33001** of FIG. **33** may be provided using tracking markers **33001a**, **33001b**, **33001c**, and **33001d** supported on array mounting frame **33011a**. Tracking markers **33001a**, **33001b**, **33001c**, and **33001d** may be detachable coupled with array mounting frame **33011a**, for example, using detachable mechanical and/or magnetic couplings, to allow attachment over/through a sterile drape as shown in FIGS. **45A**, **45B**, and **45C**. In addition, a second array mounting frame **33011b** (and respective tracking markers) may be provided for a second active marker array on an opposite side of the fixture. With two active marker arrays, tracking sensors (e.g., cameras) may track the x-ray detector from either side or both sides of the x-

ray detector.

[0267] Fixture mounting pins **38001a**, **38001b**, **38001c**, and **38001d** provide a detachable mounting with the x-ray detector as shown in FIGS. **40**, **41**, **43**, **44**, and **45A**. In some embodiments of FIGS. **38A** and **38B**, each mounting pin may include a pair of snap-fit projections, where each snap-fit projections includes a flexible stem and an outward facing head with an entrance surface at an acute angle with respect to the stem and a retention surface that is orthogonal with respect to the stem. The two snap-fit projections of a mounting pin can thus be inserted into a respective mounting socket. The entrance surfaces of the two heads facilitate entry of the mounting pin into the mounting socket and inward flexing of the respective stems. Once the mounting pin is fully inserted into the mounting socket with the two heads clear of the mounting socket opening, the two snap-fit projections spring out so that the retention surfaces of the two snap-fit projections engage an outside surface of the mounting socket to securely attach the fixture to the x-ray detector. The mounting may be released by pinching the two heads of the mounting pin so that the retention surfaces of the respective heads disengage from the outer surface of the mounting socket allowing the mounting pin to be pulled/pushed free of the mounting socket. Accordingly, each of mounting pins **38001a**, **38001b**, **38001c**, and **38001d** may be configured to detachably mate with a respective mounting socket.

[0268] In addition, the fixture may include mounting points **38015** to provide temporary mounting for a second x-ray opaque fiducial pattern in a second plane as discussed with respect to FIGS. **41** and **44**.

[0269] FIGS. **39** and **42** illustrating mounting sockets **39001a**, **39001b**, **39001c**, and **39001d** on x-ray detector **3005**. Each of the mounting sockets is configured to receive a respective mounting pin as discussed above with respect to FIGS. **38A** and **38B**. The fixture of FIGS. **38A** and **38B** can thus be detachably mounted on x-ray detector **3005** as shown in FIGS. **40**, **41**, **43**, **44**, and **45A**. In FIG. **43**, for example, each mounting pin may be released from the respective socket by pinching the head and pushing it into the respective mounting socket.

[0270] As shown in FIGS. **41** and **44**, a second x-ray opaque fiducial pattern **44001** (e.g., a pattern of BBs) may be provided in a second fiducial pattern plane **41001** that is temporarily mounted on the fixture using legs **44009** and mounting pins **44005** that detachably mate with mounting points **38015** (discussed above with respect to FIGS. **38A** and **38B**), and the second x-ray opaque fiducial pattern **44001** may be used to generate the correlation of offsets of x-ray source **3001** relative to x-ray detector **3005** as a function of an orientation of x-ray detector **3005** relative to gravity. For example, the fluoroscopic x-ray imaging system may take an air-shot image (i.e., an image without a patient present) at an orientation of the C-arm **3009** (e.g., a reference orientation of the C-arm, for example, with the x-ray detector **3005** positioned vertically above the x-ray source **3001**) with both the fixture including the first fiducial pattern and the second fiducial pattern. The air-shot image thus includes shadows corresponding to the two fiducial patterns in the different planes, and this air-shot image can be provided to the medical navigation system. For the air-shot image, the fixture can also provide information regarding an orientation of x-ray detector **3005** relative to gravity based on information from IMU **33005**, or the medical navigation system can determine an orientation of x-ray detector **3005** relative to gravity based on an orientation of a gravity vector using information received via tracking sensors (e.g., camera system) as discussed above with respect to FIGS. **23A**, **23B**, **23C**, **24A**, **24B**, **27A**, and **31**. Accordingly, the medical navigation system can generate the correlation of offsets based on the air-shot image including the shadows of the two fiducial patterns and using the information regarding the orientation of x-ray detector **3005** relative to gravity for the air-shot image. According to some other embodiments, the second x-ray opaque fiducial pattern may be temporarily provided on x-ray source **3001** as discussed above with respect to FIG. **27C**. According to some other embodiments, an optical tracking array may be coupled with x-ray source **3001** as discussed above with respect to FIG. **27C**, and the medical navigation system may determine a location/orientation of x-ray source **3001** using information

received via tracking sensors for the air-shot image and the locations/orientations of x-ray source **3001** may be used in addition to information discussed above to determine the correlation of offsets.

[0271] According to some embodiments of inventive concepts, the medical navigation system is configured to generate the correlation of offsets based on the air-shot image from the fluoroscopic imaging system with the air-shot image corresponding to one orientation of the C-arm (e.g., a reference orientation of the C-arm, for example, with the x-ray detector **3005** positioned vertically above the x-ray source **3001**). For example, the air-shot image may be generated with the x-ray opaque fiducial pattern **29000** of the fixture **38000** and the x-ray opaque fiducial pattern **44001** that is temporarily coupled with the x-ray detector between the fixture and the x-ray source as shown in FIGS. **41** and **44** so that the air-shot image includes shadows corresponding to the x-ray opaque fiducial patterns **29000** and **44001**. The medical navigation system may be configured to generate the correlation of offsets based on the air-shot image including the shadows corresponding to the x-ray opaque fiducial patterns **29000** and **44001**, and/or based on positions of the x-ray source and/or x-ray detector corresponding to respective different orientations of the C-arm, with the positions of the x-ray source being determined based on an optical tracking array coupled with the x-ray source, and/or with positions of the x-ray detector determined based on optical tracking array **33001**.

[0272] According to some other embodiments of inventive concepts, the medical navigation system is configured to generate the correlation of offsets based on an air-shot image from the fluoroscopic imaging system wherein the air-shot image corresponds to one orientation of the C-arm (e.g., a reference orientation of the C-arm, for example, with the x-ray detector **3005** positioned vertically above the x-ray source **3001**). For example, the air-shot image may be generated with the x-ray opaque fiducial pattern **29000** of the fixture **38000** and with the x-ray opaque fiducial pattern **8055** that is temporarily coupled with the x-ray source between the fixture and the x-ray source as shown in FIGS. **27C** and **31** so that the air-shot image includes shadows corresponding to the x-ray opaque fiducial patterns **29000** and **8055**. The medical navigation system may be configured to generate the correlation of offsets based on the air-shot image including the shadows corresponding to the first x-ray opaque fiducial patterns **29000** and **8055**, and/or based on positions of the x-ray source and/or x-ray detector corresponding to respective different orientations of the C-arm, with positions of the x-ray source being determined based on optical tracking array coupled with the x-ray source, and/or with positions of the x-ray detector determined based on optical tracking array **33001**.

[0273] Once the correlation of offsets has been determined, the second x-ray opaque fiducial pattern (on the fixture or on x-ray source **3001**) and/or the tracking array on the x-ray source **3001** may be removed before performing a medical procedure. The correlation of offsets may be stored in the medical navigation system and used for registration, or the correlation of offsets may be saved in memory at the fixture. If the correlation of offsets is saved at the fixture, the fixture can determine the orientation of x-ray detector **3005** relative to gravity (using output from IMU **33005**) for each patient image during a procedure, determine the offset for the patient image based on the orientation relative to gravity and the correlation of offsets for each patient image, and transmit an indication of the offset to the navigation system for each patient image. If the correlation of offsets is saved at the navigation system, the fixture can determine the orientation of x-ray detector **3005** relative to gravity (using output from IMU **33005**) for each patient image during a procedure and transmit the indication of the orientation relative to gravity to the medical navigation system for each patient image, and the medical navigation system can determine the offset for the patient image based on the orientation relative to gravity and the correlation of offsets for each patient image. Either way, the medical navigation system can perform registration based on the air-show images and based on the respective offsets.

[0274] As shown in FIGS. **45A**, **45B**, and **45C**, once the fixture is mounted on x-ray detector **3005** and any temporary fiducial/tracking arrays have been removed, a sterile drape **33015** may be provided over the fixture and x-ray detector **3005**. For example, because of electronic components

included in the fixture, it may be difficult to sufficiently sterilize the fixture for surgery, and sterile drape **33015** may thus be required. Once sterile drape **33015** has been provided, sterile tracking markers **33001a**, **33001b**, **33001c**, and **33001d** may be coupled with mounting points **44002a**, **44002b**, **44002c**, and **44002d** of array mounting frame **33001a**, either over or through the sterile drape. For example, the tracking markers may be magnetically mounted on respective mounting points without penetrating the sterile drape, or a snap fit pin of each tracking marker may penetrate the sterile drape to provide a snap fit mounting on a respective mounting point.

[0275] According to some embodiments, an optical surveillance marker **8091** may be provided on C-arm **3009** and/or on x-ray detector **3005** as discussed above with respect to FIG. **31**. When the medical navigation system performs registration according to some embodiments, the optical surveillance marker **8091** can be used to validate the patient images.

[0276] FIG. **34** is a block diagram illustrating elements of a fixture controller **34000** for the detector fixture discussed above with respect to FIG. **33**. As shown, the fixture controller **34000** may include processor circuitry **34007** (also referred to as a processor) coupled with inertial measurement unit (IMU) circuitry **33005** (also referred to as an inertial measurement unit or IMU), x-ray presence sensor circuitry **33019** (also referred to as an x-ray presence sensor/detector), marker array interface circuitry **33051** (also referred to as a marker array interface), memory circuitry **33009A** (also referred to as memory), and communication interface circuitry **33009B** (also referred to as a communication interface). The memory circuit **33009A** may include computer readable program code that when executed by the processor circuit **34007** causes the processor circuit to perform operations according to embodiments disclosed herein. According to other embodiments, processor circuit **34007** may be defined to include memory so that a separate memory circuit is not required.

[0277] FIG. **35** is a block diagram illustrating elements of medical navigation system controller **35000** (e.g., implemented within computer **408** of the robotic system of FIG. **5**). As shown, the medical navigation system controller **35000** may include processor circuitry **35007** (also referred to as a processor) coupled with input interface circuitry **35001** (also referred to as an input interface), output interface circuitry **35003** (also referred to as an output interface), control interface circuitry **35005** (also referred to as a control interface), memory circuitry **35009** (also referred to as a memory), and communication interface circuitry **35051** (also referred to as a communication interface). The memory circuitry **35009** may include computer readable program code that when executed by processor circuitry **35007** causes the processor circuitry to perform operations according to embodiments disclosed herein. According to other embodiments, processor circuitry **35007** may be defined to include memory so that separate memory circuitry is not required. Processor **35007** may use communication interface **35051** to receive information regarding an offset of an x-ray source of the fluoroscopic x-ray imaging system relative to an x-ray detector of the fluoroscopic x-ray imaging system. The communication interface **35051**, for example, may receive such information from the fixture of FIGS. **34** and **35** using wired or wireless (e.g., WiFi, Bluetooth, etc.) communication.

[0278] As discussed herein, operations of controlling a medical navigation system according to some embodiments of the present disclosure may be performed by controller **35000** including processor **35007**, input interface **35001**, output interface **35003**, control interface **35005**, and/or communication interface **35051**. For example, processor **35007** may receive user input through input interface **35001**, and such user input may include user input received through foot pedal **544**, tablet **546**, a touch sensitive interface of display **110/304**, etc. Processor **35007** may also receive position sensor input from tracking subsystem **532** and/or cameras **200** through input interface **35001**. Processor **35007** may provide output through output interface **35003**, and such output may include information to render graphic/visual information on display **110/304** and/or audio output to be provided through speaker **536**. Processor **35007** may provide robotic control information through control interface **35005** to motion control subsystem **506**, and the robotic control

information may be used to control operation of a robotic actuator (such as robot arm **104/306-308/604**, also referred to as a robotic arm), and/or end-effector **112/602**. Processor **35007** may also use communication interface **35051** to receive information regarding an offset of an x-ray source of a fluoroscopic x-ray imaging system relative to an x-ray detector of the fluoroscopic x-ray imaging system. The communication interface **35051**, for example, may receive such information from the fixture of FIGS. **33** and **34** using wired or wireless (e.g., WiFi, Bluetooth, etc.) communication. [0279] Operations of a fixture for a fluoroscopic x-ray imaging system including a C-arm will now be discussed with reference to the flow charts of FIGS. **36A** and **36B** according to some embodiments of inventive concepts. For example, modules may be stored in memory **33009A** of FIG. **34**, and these modules may provide instructions so that when the instructions of a module are executed by processor **34007**, processor **34007** performs respective operations of the flow chart of FIG. **36A** and/or **36B**.

[0280] At block **36001** of FIG. **36A**, a correlation between different orientations of the x-ray detector **3005** (and/or C-arm **3009**) relative to gravity and respective offsets of the x-ray source **3001** relative to the x-ray detector **3005** may be provided in memory **33009A** for the fluoroscopic x-ray imaging system. According to some embodiments, a single correlation may be provided in memory **33009A** at block **36001** for a single fluoroscopic imaging system, and in such embodiments, the fixture may be used with a single fluoroscopic x-ray imaging system for which the correlation applies. The correlation may be provided, for example, based on one or more calibrations as discussed above.

[0281] According to some other embodiments discussed below with respect to FIG. **36B**, a plurality of correlations may be provided in memory **33009A** for respective different fluoroscopic x-ray imaging systems so that the same fixture may be used with different fluoroscopic x-ray imaging systems. According to such embodiments at block **36001A**, a plurality of correlations between different orientations of C-arms **3009** and/or x-ray detectors **3005** and respective offsets of x-ray sources **3001** relative to x-ray detectors **3005** for different fluoroscopic x-ray imaging systems are provided in memory **33009A**, and each of the plurality of correlations is associated with a different identification for the different fluoroscopic x-ray imaging systems. Accordingly, a different correlation may be provided in memory **33009A** for each of the fluoroscopic imaging systems with which the fixture will be used, allowing one fixture to be used with different fluoroscopic x-ray imaging systems. At block **36001B**, processor **34007** receives (e.g., through communication interface **33009B**) an identification corresponding to the fluoroscopic imaging system with which the fixture will be used. For example, a smartphone (or other electronic device) may be used to read/capture an identification (e.g., a bar code, a QR code, an NFC tag, an RFID tag, etc.) on the fluoroscopic x-ray imaging system with which the fixture will be used, and the identification may be received by processor **34007** through communication interface **33009B** via wireless communication from the smartphone or other electronic device, or processor **34007** may receive the identification directly through communication interface **33009B** (e.g., by reading a Near Field Communication NFC tag or a Radio Frequency Identification RFID tag). At block **36001C**, processor **34007** may provide the correlation of block **36001** by selecting the correlation corresponding to the identification received at block **36001B**. The fixture may thus include an attachment mount configured to provide a detachable mechanical coupling with each of the different fluoroscopic x-ray imaging systems, and upon mounting the fixture on one of the fluoroscopic x-ray imaging systems using the attachment mount, the identification of that x-ray imaging system can be used to select the correlation for that fluoroscopic x-ray imaging system. Moreover, the fixture may include an x-ray opaque fiducial pattern in a single plane, and a mounting structure (e.g., mounting pins **38001a-d** of FIGS. **38A** and **38B**) may be configured to provide the detachable mechanical coupling with x-ray detector **3005** so that the fiducial pattern is on a face of the x-ray detector between the x-ray detector **3001** and the x-ray source **3005** when the fixture is coupled to the x-ray detector using the mounting structure.

[0282] According to some embodiments, the correlation of block **36001** (or each correlation of block **36001A**) may include a table in memory **33009A** relating the different orientations of the x-ray detector **3005** (and/or C-arm **3009**) relative to gravity and the respective offsets, a formula in memory **33009A** relating the different orientations of the x-ray detector **3005** (and/or C-arm **3009**) relative to gravity and the respective offsets, and/or a model in memory **33009A** relating the different orientations of the x-ray detector **3005** (and/or C-arm **3009**) relative to gravity and the respective offsets. If the fixture is provided with correlations for different fluoroscopic x-ray imaging systems, a different table, formula, and/or model may be provided in memory **33009A** for each of the fluoroscopic x-ray imaging systems.

[0283] At block **36005**, processor **34007** determines whether x-ray emission from x-ray source **3001** to x-ray detector **3005** is present/detected based on information from x-ray presence sensor **33019**. By providing x-ray presence sensor **33019** on face **3006** of detector **3005**, output from x-ray presence sensor **33019** can be used to determine when a fluoroscopic image/shot is being taken, and thus, when offset information is needed.

[0284] According to some embodiments at block **36009**, processor **34007** determines an orientation of the C-arm **3009** and/or detector **3005** responsive to detecting an x-ray emission from the x-ray source toward the x-ray detector at block **36005**. For example, the orientation of the C-arm and/or detector **3005** may be determined relative to gravity based on an output from inertial measurement unit IMU **33005**. By determining the orientation responsive to detecting the x-ray emission, the orientation is determined at the time of taking the fluoroscopic x-ray image/shot. Moreover, the orientation of the C-arm and/or detector **3005** may be determined/defined with respect to a reference orientation of the C-arm and/or detector **3005**, where the reference orientation of the C-arm and/or detector **3005** is defined as the orientation of the C-arm and/or detector **3005** with the x-ray detector vertically above the x-ray source.

[0285] According to some embodiments at block **36011**, processor **34007** determines an offset of the x-ray source **3001** relative to the x-ray detector **3005** responsive to detecting an x-ray emission from the x-ray source **3001** toward the x-ray detector **3005** at block **36005**. For example, the offset of the x-ray source **3001** relative to the x-ray detector **3005** may be determined based on the orientation of the x-ray detector **3005** (and/or C-arm **3009**) from block **36009**, based on the orientation of the x-ray detector **3005** (and/or C-arm **3009**) from block **36009** and based on the correlation in memory **33009A** from block **36001**, and/or based on the orientation of the x-ray detector **3005** (and/or C-arm **3009**) from block **36009** and based on the correlation in memory **33009A** being associated with the fluoroscopic x-ray imaging system based on the identification as discussed above with respect to blocks **36001A-C**. The offset of the x-ray source relative to the x-ray detector may be determined/defined, for example, using a coordinate system of the x-ray detector where an x-axis and a y-axis are defined in a plane of the x-ray detector and a z axis is defined perpendicular to the plane of the x-ray detector.

[0286] According to some embodiments at block **36015**, processor **34007** activates a tracking marker through marker array interface **33051** responsive to detecting the x-ray emission from block **36005**. For example, the tracking marker may be one of a plurality of light emitting diodes (LEDs) of tracking marker array **33001** and activating the tracking marker may cause the plurality of LEDs to turn on, and/or the tracking marker may be a moveable tracking marker and activating the tracking marker may cause the moveable tracking marker to move from a respective first position to a respective second position. As discussed above with respect to FIG. **33**, marker array **33001** may be coupled to array mount **33011** through sterile drape **33015** that covers the fixture and at least a portion of detector **3005**. Marker array **33001** (or a tracking marker thereof) may thus be activated by processor **34007** using interface **33051** (e.g., through array mount **33011**), for example, using electrical coupling, inductive coupling, mechanical coupling, radio coupling, electromagnetic coupling, magnetic coupling, etc.

[0287] According to some embodiments, tracking marker/markers may be activated at block **36015**

by turning on one or more light emitting diode (LED) tracking markers responsive to detecting x-ray emission at block **36005**. According to some other embodiments, a fixed/passive array of tracking markers and a moveable tracking marker may be provided for the fixture, and the moveable tracking marker may be activated at block **36015** by moving the moveable tracking marker relative to the array of tracking markers responsive to detecting x-ray emissions at block **36005**.

[0288] According to some embodiments at block **36017**, processor **34007** provides an indication of the offset of the x-ray source from block **36011** through communication interface **33009B** to the medical navigation system. The offset, for example, may be provided through communication interface **33009B** to the medical navigation system via a wired coupling or via a wireless coupling (e.g., via Wifi, Bluetooth, etc.). The medical navigation system can thus use the indication of the offset to provide registration as discussed below.

[0289] According to some other embodiments, the correlation between x-ray detector (and/or C-arm) orientations relative to gravity and offsets may be provided at the medical navigation system (instead of the fixture). In such embodiments, processor **34007** may provide an indication of the x-ray detector **3005** (and/or C-arm) orientation relative to gravity through communication interface **33009B** to the medical navigation (via wired or wireless coupling, e.g., via Wifi, Bluetooth, etc.) responsive to detecting x-ray emission, and the medical navigation system may use the x-ray detector **3005** (and/or C-arm) orientation relative to gravity and the correlation between x-ray detector **3005** (and/or C-arm) orientations and offsets to determine/provide the offset of the x-ray source relative to the x-ray detector. In either embodiment, processor **34007** may provide information (e.g., the detector/C-arm orientation or the offset) based on the x-ray detector **3005** (and/or C-arm orientation) relative to gravity to the medical navigation system for use by the medical navigation system to perform registration. Because the C-arm and the x-ray detector are connected and/or integrated, an orientation of the x-ray detector **3005** may refer to an orientation of the C-arm, and/or an orientation of the C-arm **3009** may refer to an orientation of the x-ray detector **3005**.

[0290] Operations of a medical navigation system will now be discussed with reference to the flow chart of FIG. **37** according to some embodiments of inventive concepts. For example, modules may be stored in memory **35009** of FIG. **35**, and these modules may provide instructions so that when the instructions of a module are executed by processor **35007**, processor **35007** performs respective operations of the flow chart of FIG. **35**.

[0291] According to some embodiments at block **37005**, processor **35007** receives information defining a first patient image from a fluoroscopic x-ray imaging system through communication interface **35051**, wherein the fluoroscopic x-ray imaging system includes a C-arm **3009**, an x-ray source **3001** at a first end of the C-arm, and an x-ray detector **3005** at a second end of the C-arm. The first patient image may include shadows corresponding to an x-ray opaque fiducial pattern of the fixture.

[0292] According to some embodiments at block **37009**, processor **35007** receives an indication of a first offset of the x-ray source **3001** relative to the x-ray detector **3005**, wherein the first offset is associated with the first patient image, and wherein the indication of the first offset is received through communication interface **35051**.

[0293] According to some embodiments, receiving the information defining the first patient image at block **37005** and receiving the indication of the first offset at block **37009** may include receiving a digital file (e.g., a Digital Imaging and Communication in Medicine DCOM file) including the information defining the first patient image and the indication of the first offset. According to some other embodiments, the first offset may be received separately from the information defining the first patient image. For example, the first offset may be received via a wireless coupling, such as a WiFi coupling, a Bluetooth coupling, etc.

[0294] According to some embodiments at block **37011**, processor **35007** determines a first

location (also referred to as position) of the x-ray detector **3005** and/or x-ray source **3001** based on a first location (also referred to as position) of tracking marker array **33001** on x-ray detector **3005** determined using tracking information from tracking sensors (e.g., tracking cameras **200**), with the first location being associated with the first patient image. For example, the first location of the x-ray detector may be determined based on a location of a tracking marker array **33001** determined using the tracking information from the tracking sensors, and/or the first location of the x-ray detector may be determined responsive to detecting a first activation of a tracking marker (e.g., one or more of the markers of array **33001**) using the tracking information from the tracking sensors. Activation may be detected, for example, based on detecting movement of a marker(s) and/or based on detecting active illumination of a marker(s).

[0295] According to some embodiments at block **37012**, processor **35007** determines a first location (also referred to as position) of a surveillance marker **8091** using tracking information from the tracking sensors, with the first location of the surveillance marker **8091** being associated with the first patient image. At block **37014**, processor **35007** may determine that the first location of the x-ray detector and/or x-ray source is valid based on the first location of the surveillance marker **8091** relative to the tracking marker array at the first location of the tracking marker array. If processor **35007** does not determine that the first location of the x-ray detector is valid, validation based on the first patient image, the first offset, and/or the first location of the x-ray detector may be blocked and/or another first patient image (and associated information, e.g., offset) may be requested.

[0296] According to some embodiments at block **37015**, processor **35007** receives information defining a second patient image from the fluoroscopic x-ray imaging system through communication interface **35051**, with the first and second patient images being different.

[0297] According to some embodiments at block **37019**, processor **35007** receives an indication of a second offset of the x-ray source **3001** relative to the x-ray detector **3005**, wherein the second offset is associated with the second patient image, wherein the first and second offsets are different, and wherein the indication of the second offset is received through communication interface **35051**.

[0298] According to some embodiments, receiving the information defining the second patient image at block **37005** and receiving the indication of the second offset at block **37009** may include receiving a digital file (e.g., a Digital Imaging and Communication in Medicine DCOM file) including the information defining the second patient image and the indication of the second offset. According to some other embodiments, the second offset may be received separately from the information defining the second patient image. For example, the second offset may be received via a wireless coupling, such as a WiFi coupling, a Bluetooth coupling, etc.

[0299] According to some embodiments at block **37021**, processor **35007** determines a second location (also referred to as position) of the x-ray detector **3005** and/or x-ray source **3001** based on a second location (also referred to as position) of tracking marker array **33001** on x-ray detector **3005** determined using tracking information from tracking sensors (e.g., tracking cameras **200**), with the second location being associated with the second patient image, and with the first and second patient images being different. For example, the second location of the x-ray detector may be determined based on a location of a tracking marker array **33001** determined using the tracking information from the tracking sensors, and/or the second location of the x-ray detector may be determined responsive to detecting a second activation of a tracking marker (e.g., one or more of the markers of array **33001**) using the tracking information from the tracking sensors. Activation may be detected, for example, based on detecting movement of a marker(s) and/or based on detecting active illumination of a marker(s).

[0300] According to some embodiments at block **37022**, processor **35007** determines a second location (also referred to as position) of the surveillance marker **8091** using tracking information from the tracking sensors, with the second location of the surveillance marker **8091** being associated with the second patient image. At block **37014**, processor **35007** may determine that the

second location of the x-ray detector and/or x-ray source is valid based on the second location of the surveillance marker **8091** relative to the tracking marker array at the second location of the tracking marker array. If processor **35007** does not determine that the second location of the x-ray detector is valid, validation based on the second patient image, the second offset, and/or the second location of the x-ray detector may be blocked and/or another second patient image (and associated information, e.g., offset) may be requested.

[0301] According to some embodiments at block **37025**, processor **35007** provides a registration between a tracking coordinate system for a physical space monitored by the tracking sensors (e.g., tracking cameras **200**) and an image coordinate system for the first and second patient images from the fluoroscopic x-ray imaging system, wherein the registration is provided based on the indications of the first and second offsets. For example, the registration between the tracking coordinate system and the image coordinate system may be provided based on the first and second offsets, based on the first and second locations of the x-ray detector, based on the correlation of offsets of x-ray source relative to the x-ray detector as a function of an orientation of the x-ray detector relative to gravity, and/or based on the shadows in the first and second patient images. According to some embodiments, processor **35007** may determine a first location of the x-ray source based on the first location of the x-ray detector (e.g., determined using information regarding marker array **33001** received through optical tracking/camera system) and the first offset, processor **35007** may determine a second location of the x-ray source based on the second location of the x-ray detector (e.g., determined using information regarding marker array **33001** received through optical tracking/camera system) and the second offset, and the registration may be provided based on one or more of the correlation of offsets, based on the first offset, based on the second offset, based on the first and second shadows, based on the first location of the x-ray detector, based on the second location of the x-ray detector, based on the first location of the x-ray source, and/or based on the second location of the x-ray source. Moreover, registration may be provided responsive to determining that the first and second locations of the x-ray detector are valid at blocks **37014** and **37024**.

[0302] For example, the registration may be provided responsive to detecting activation of a tracking marker (e.g., one or more of the markers of array **33001**) using the tracking information from the tracking sensors. Activation may be detected, for example, based on detecting movement of the marker(s) and/or based on detecting active illumination of the marker(s). For example, the registration may be provided responsive to detecting activation of a tracking marker (e.g., one or more of the markers of array **33001**) using the tracking information from the tracking sensors. Activation may be detected, for example, based on detecting movement of the marker(s) and/or based on detecting active illumination of the marker(s).

[0303] According to some embodiments at block **37029**, processor **35007** may control the robotic actuator to move the end-effector to a target trajectory relative to the patient based on the first and second patient images from the fluoroscopic x-ray imaging system and based on the registration between the tracking coordinate system and the image coordinate system. For example, controlling the robotic actuator may include controlling the robotic actuator to move the end-effector to the target trajectory relative to the patient based on the first and second patient images from the fluoroscopic x-ray imaging system, based on the registration between the tracking coordinate system and the image coordinate system, and based on a location of the end-effector determined using tracking information from the tracking sensors. According to some other embodiments, processor **35007** may render an image on a display based on the first and second patient images of blocks **37005** and **37015** and based on the registration. In such embodiments, processor **35007** may render the image to further include a portion thereof based on an image received through the optical tracking/camera system and/or to further include a portion thereof based on a virtual element (e.g., a virtual screw or other medical device) that may be used for planning.

[0304] As discussed above with respect to FIG. 37, the medical navigation system may receive

respective offsets from the fixture at blocks **37009** and **37019** so that the correlation between x-ray detector **3005** (and/or C-arm) orientations relative to gravity and offsets may be provided at the fixture and the offset may be determined at the fixture. According to some other embodiments, the correlation between x-ray detector **3005** (and/or C-arm) orientations and offsets may be provided in memory **35009** at the medical navigation system, and the medical navigation system may instead receive (through communication interface **35051**) indications of the x-ray detector **3005** (and/or C-arm) orientation relative to gravity at blocks **37009** and **37019**. In such embodiments, processor **35007** may determine the respective offsets using the indications of the x-ray detector **3005** (and/or C-arm) orientations and the correlation between x-ray detector **3005** (and/or C-arm) orientations and offsets to determine/provide the offset of the x-ray source relative to the x-ray detector. In either embodiment, processor **35007** may use information (e.g., the C-arm/detector orientation or the offset) received from the fixture to determine/provide the offsets that are used to perform registration.

[0305] According to still other embodiments, the fixture may not provide any information regarding an orientation of the C-arm/detector or an offset. In such embodiments, processor **35007** may determine orientations of the C-arm/detector at blocks **37009** and **37019** for respective images using a gravity vector on the C-arm and/or x-ray detector as discussed above with respect to FIGS. **23A**, **23B**, **23C**, **24A**, **24B**, **27A**, and **31**. As discussed above, the gravity vector includes at least two suspended tracking markers, and processor **35007** can determine the orientations of the c-arm/detector relative to gravity by determining positions of the gravity vector tracking markers based on information received through the tracking sensors/cameras at the time of the respective patient image. In such embodiments, the fixture may signal the timing of an image/shot by activating one or more markers of the active tracking marker array, and responsive to detecting activation of the one or more markers of the active tracking marker array, processor **35007** may use the gravity vector to determine the orientation of the c-arm/detector for the respective image. Where the gravity vector includes fixed tracking markers **4005A**, **4005B**, **4005C**, and **4005D** and moveably suspended tracking markers **4001A** and **4001B**, processor **35007** can use the fixed tracking markers to determine a location of the x-ray detector, and processor **35007** can use the moveably suspended markers to determine the orientation relative to gravity.

[0306] In the above-description of various embodiments of present inventive concepts, it is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of present inventive concepts. Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which present inventive concepts belong. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0307] When an element is referred to as being “connected”, “coupled”, “responsive”, or variants thereof to another element, it can be directly connected, coupled, or responsive to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected”, “directly coupled”, “directly responsive”, or variants thereof to another element, there are no intervening elements present. Like numbers refer to like elements throughout. Furthermore, “coupled”, “connected”, “responsive”, or variants thereof as used herein may include wirelessly coupled, connected, or responsive. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Well-known functions or constructions may not be described in detail for brevity and/or clarity. The term “and/or” includes any and all combinations of one or more of the associated listed items.

[0308] It will be understood that although the terms first, second, third, etc. may be used herein to

describe various elements/operations, these elements/operations should not be limited by these terms. These terms are only used to distinguish one element/operation from another element/operation. Thus, a first element/operation in some embodiments could be termed a second element/operation in other embodiments without departing from the teachings of present inventive concepts. The same reference numerals or the same reference designators denote the same or similar elements throughout the specification.

[0309] As used herein, the terms “comprise”, “comprising”, “comprises”, “include”, “including”, “includes”, “have”, “has”, “having”, or variants thereof are open-ended, and include one or more stated features, integers, elements, steps, components or functions but does not preclude the presence or addition of one or more other features, integers, elements, steps, components, functions or groups thereof. Furthermore, as used herein, the common abbreviation “e.g.”, which derives from the Latin phrase “*exempli gratia*,” may be used to introduce or specify a general example or examples of a previously mentioned item, and is not intended to be limiting of such item. The common abbreviation “i.e.”, which derives from the Latin phrase “*id est*,” may be used to specify a particular item from a more general recitation.

[0310] Example embodiments are described herein with reference to block diagrams and/or flowchart illustrations of computer-implemented methods, apparatus (systems and/or devices) and/or computer program products. It is understood that a block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, can be implemented by computer program instructions that are performed by one or more computer circuits. These computer program instructions may be provided to a processor circuit of a general purpose computer circuit, special purpose computer circuit, and/or other programmable data processing circuit to produce a machine, such that the instructions, which execute via the processor of the computer and/or other programmable data processing apparatus, transform and control transistors, values stored in memory locations, and other hardware components within such circuitry to implement the functions/acts specified in the block diagrams and/or flowchart block or blocks, and thereby create means (functionality) and/or structure for implementing the functions/acts specified in the block diagrams and/or flowchart block(s).

[0311] These computer program instructions may also be stored in a tangible computer-readable medium that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable medium produce an article of manufacture including instructions which implement the functions/acts specified in the block diagrams and/or flowchart block or blocks. Accordingly, embodiments of present inventive concepts may be embodied in hardware and/or in software (including firmware, resident software, micro-code, etc.) that runs on a processor such as a digital signal processor, which may collectively be referred to as “circuitry,” “a module” or variants thereof.

[0312] It should also be noted that in some alternate implementations, the functions/acts noted in the blocks may occur out of the order noted in the flowcharts. For example, two blocks shown in succession may in fact be executed substantially concurrently or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved. Moreover, the functionality of a given block of the flowcharts and/or block diagrams may be separated into multiple blocks and/or the functionality of two or more blocks of the flowcharts and/or block diagrams may be at least partially integrated. Finally, other blocks may be added/inserted between the blocks that are illustrated, and/or blocks/operations may be omitted without departing from the scope of inventive concepts. Moreover, although some of the diagrams include arrows on communication paths to show a primary direction of communication, it is to be understood that communication may occur in the opposite direction to the depicted arrows.

[0313] Although several embodiments of inventive concepts have been disclosed in the foregoing specification, it is understood that many modifications and other embodiments of inventive concepts will come to mind to which inventive concepts pertain, having the benefit of teachings

presented in the foregoing description and associated drawings. It is thus understood that inventive concepts are not limited to the specific embodiments disclosed hereinabove, and that many modifications and other embodiments are intended to be included within the scope of the appended claims. It is further envisioned that features from one embodiment may be combined or used with the features from a different embodiment(s) described herein. Moreover, although specific terms are employed herein, as well as in the claims which follow, they are used only in a generic and descriptive sense, and not for the purposes of limiting the described inventive concepts, nor the claims which follow. The entire disclosure of each patent and patent publication cited herein is incorporated by reference herein in its entirety, as if each such patent or publication were individually incorporated by reference herein. Various features and/or potential advantages of inventive concepts are set forth in the following claims.

Claims

1. A method of registration for a fluoroscopic x-ray imaging system, wherein the fluoroscopic x-ray imaging system includes a C-arm, an x-ray source at a first end of the C-arm, and an x-ray detector at a second end of the C-arm, the method comprising: providing a fixture for the fluoroscopic x-ray imaging system, the fixture including an x-ray opaque fiducial pattern in a single plane, wherein the fixture is configured to be coupled to the C-arm so that the x-ray opaque fiducial pattern is on an x-ray beam path between the x-ray detector and the x-ray emitter, receiving, by a medical navigation system, a first patient image from the fluoroscopic imaging system corresponding to a first orientation of the C-arm with the first patient image including first shadows corresponding to the x-ray opaque fiducial pattern; receiving, by a medical navigation system, a second patient image from the fluoroscopic imaging system corresponding to a second orientation of the C-arm with the second patient image including second shadows corresponding to the x-ray opaque fiducial pattern; and providing a registration, by the medical navigation system, between a tracking coordinate system for a physical space monitored by tracking cameras and an image coordinate system for the first patient image and the second patient image based on a correlation of offsets of the x-ray source relative to the x-ray detector.
2. The method of claim 1, wherein the x-ray opaque fiducial pattern comprises plurality of BB fiducials residing in the single plane.
3. The method of claim 2, wherein the plurality of BB fiducials includes first, second, third, and fourth clusters of BB fiducials, wherein the first cluster of BB fiducials is spaced apart from the third cluster of BB fiducials along a first axis, wherein the second cluster of BB fiducials is spaced apart from the fourth cluster of BB fiducials along a second axis, and wherein the first axis is orthogonal with respect to the second axis.
4. The method of claim 1, wherein the x-ray opaque fiducial pattern comprises first, second, third, and fourth x-ray opaque fiducial segments that extend in different directions from a common point, wherein the first x-ray opaque fiducial segment has a first break at a first distance from the common point, wherein the second x-ray opaque fiducial segment has a second break at a second distance from the common point greater than the first distance, and wherein the third x-ray opaque fiducial segment has a third break at a third distance from the common point greater than the second distance.
5. The method of claim 1, wherein providing the registration includes providing the registration based on the correlation, based on a first orientation of the x-ray detector relative to gravity associated with the first orientation of the C-arm, and based on a second orientation of the x-ray detector relative to gravity associated with the second orientation of the C-arm.
6. The method of claim 5, wherein the fixture includes an inertial measurement unit (IMU) configured to detect an orientation of the x-ray detector relative to gravity, and wherein the first orientation of the x-ray detector relative to gravity is determined based on a first output from the

inertial measurement unit with the C-arm in the first orientation, and wherein the second orientation of the x-ray detector relative to gravity is determined based on a second output from the inertial measurement unit with the C-arm in the second orientation.

7. The method of claim 5, the fixture further including a gravity vector coupled with the x-ray detector, wherein the gravity vector includes a pair of movably suspended optical tracking markers, wherein the first orientation of the x-ray detector relative to gravity is determined based on first positions of the pair of movably suspended optical tracking markers with the C-arm in the first orientation, and wherein the second orientation of the x-ray detector relative to gravity is determined based on second positions of the pair of movably suspended optical tracking markers with the C-arm in the second orientation.

8. The method of claim 1, wherein the fixture includes a plurality of mounting pins, wherein each of the mounting pins is configured to detachably mate with a respective one of the mounting sockets.

9. The method of claim 1, further comprising generating, by the medical navigation system, the correlation of offsets based on an air-shot image from the fluoroscopic imaging system wherein the air-shot image corresponds to one orientation of the C-arm.

10. The method of claim 9, wherein the x-ray opaque fiducial pattern is a first x-ray opaque fiducial pattern, wherein a second x-ray opaque fiducial pattern is temporarily coupled with the x-ray detector between the fixture and the x-ray source so that the air-shot image includes shadows corresponding to the first x-ray opaque fiducial pattern and the second x-ray opaque fiducial pattern, and the method further comprising: generating, by the medical navigation system, the correlation of offsets based on the air-shot image including the shadows corresponding to the first x-ray opaque fiducial pattern and the second x-ray opaque fiducial pattern.

11. The method of claim 9, wherein the x-ray opaque fiducial pattern is a first x-ray opaque fiducial pattern, wherein a second x-ray opaque fiducial pattern is temporarily coupled with the x-ray source between the fixture and the x-ray source so that the air-shot image includes shadows corresponding to the first x-ray opaque fiducial pattern and the second x-ray opaque fiducial pattern, and the method further comprising: generating, by the medical navigation system, the correlation of offsets based on the air-shot image including the shadows corresponding to the first x-ray opaque fiducial pattern and the second x-ray opaque fiducial pattern.

12. The method of claim 9, the method further comprising: generating, by the medical navigation system, the correlation of offsets based on the air-shot image and based on positions of the x-ray source corresponding to respective different orientations of the C-arm, wherein the positions of the x-ray source are determined based on an optical tracking array coupled with the x-ray source.

13. The method of claim 1, wherein the fixture further includes: an optical tracking array coupled with the fixture, and an optical surveillance marker coupled with the x-ray detector and/or the x-ray source, the method further comprising: determining, by the medical navigation system, a first position of the x-ray detector and/or the x-ray source associated with the first patient image based on the optical tracking array; validating, by the medical navigation system, the first position of the x-ray detector and/or the x-ray source based on a first position of the optical surveillance marker associated with the optical tracking array, determine a second position of the x-ray detector and/or the x-ray source associated with the second patient image based on the optical tracking array; validating, by the medical navigation system, the second position of the x-ray detector and/or the x-ray source based on a second position of the optical surveillance marker associated with the optical tracking array; and providing the registration includes being based on the first and second shadows, based on the first position of the x-ray detector and/or the x-ray source, based on validating the first position of the x-ray detector and/or the x-ray source, based on the second position of the x-ray detector and/or the x-ray source, and based on validating the second position of the x-ray detector and/or the x-ray source.

14. A method of registration for a fluoroscopic x-ray imaging system, wherein the fluoroscopic x-

ray imaging system includes a C-arm, an x-ray source at a first end of the C-arm, and an x-ray detector at a second end of the C-arm, the method comprising: providing a fixture for the fluoroscopic x-ray imaging system, the fixture including: a processor, and memory coupled with the processor, wherein the memory includes instructions that are executable by the processor; detecting, by the fixture, an x-ray emission from the x-ray source toward the x-ray detector; determining, by the fixture, an offset of the x-ray source relative to the x-ray detector, responsive to detecting the x-ray emission, and providing, by the fixture, an indication of the offset of the x-ray source to a medical navigation system.

15. The method of claim 14, further comprising: determining, by the fixture, an orientation of the x-ray detector responsive to detecting the x-ray emission, wherein the offset of the x-ray source relative to the x-ray detector is determined based on the orientation of the x-ray detector.

16. The method of claim 15, further comprising: providing, by the fixture, in memory a correlation between different orientations of the x-ray detector and respective offsets of the x-ray source relative to the x-ray detector, wherein the offset of the x-ray source relative to the x-ray detector is determined based on the orientation of the x-ray detector and based on the correlation in memory.

17. The method of claim 14 wherein the fixture further includes: an x-ray opaque fiducial pattern in a single plane; and a mounting structure configured to provide a detachable mechanical coupling between the fixture and the x-ray detector of the fluoroscopic x-ray imaging system so that the fiducials are on a face of the x-ray detector between the x-ray detector and the x-ray source when the fixture is coupled to the x-ray detector using the mounting structure.
